

9 Lemon sole in the North Sea, Skagerrak, Kattegat, and eastern English Channel

lem.27.3a47d – *Microstomus kitt* in Subarea 4 and divisions 3.a and 7.d

9.1 General

The assessment of North Sea lemon sole (*Microstomus kitt*) was subject to a benchmark during the winter of 2017–18 (ICES, 2018). In summary, the benchmark concluded the following:

- There were insufficient age samples submitted to InterCatch to allow for a full age-structured catch-based assessment. InterCatch collation was therefore conducted on the basis of length.
- Age-structured survey indices were developed using GAM estimation (Berg *et al.* 2014), for Q1 (IBTS; ages 1–5, years 2007–present) and Q3 (IBTS and BTS; ages 1–9, years 2005–present). Only ages 2–5 for the Q1 survey were used in the assessment, due to very low sample sizes for age-1 lemon sole in the Q1 IBTS survey.
- Maturity-at-age was fixed through time (based on IBTS Q1 samples), while weights-at-age were based on smoothly-varying observations from both IBTS Q1 and Q3.
- The stock assessment model used for the basis of the advice was SURBAR (Needle, 2015), including *ad hoc* adjustments for the observed low catchability of the available surveys for age 1 and 2 lemon sole.
- The advice was based on the DLS 3.2 rule, applied to relative SSB estimates provided by SURBAR.
- Stock status in relation to F_{MSY} proxies was evaluated using a suite of length-based indicators (LBIs).

Stipulations 1-3 and 6 have been followed in this year's WGNSSK update assessment. However, while SURBAR is still used as an exploratory assessment method, the basis for the advice has now been changed to the new WKLIFE X DLS process (ICES, 2022) applied to the SURBAR estimated SSB as stock size indicator. The empirical biomass index from the IBTS Q1 survey alone is not considered suitable anymore due to low biological sampling in IBTS Q1 in 2023.

Since 2020, advice on lemon sole has been requested on an annual basis. The outcome of the current assessment will be used to provide new catch advice for 2026.

9.1.1 Biology and ecosystem aspects

Lemon sole is a commercially important flatfish that is found in the shelf waters of the North Atlantic from the White Sea and Iceland southwards to the Bay of Biscay. Lemon sole spawn for a lengthy period in the North Sea, starting as early as April in the north and ending as late as November in the south (Rae, 1965). In the western English Channel, lemon sole spawn in April and May (Jennings *et al.*, 1993). In the English Channel, investigations of habitat association for plaice, sole and lemon sole have indicated that distribution is restricted to a few sites and that lemon soles appear to prefer sandy and gravely strata, living deeper, at higher salinities and lower temperatures than plaice or sole (Hinz *et al.*, 2006). Lemon sole feed on small invertebrates, mainly polychaete worms, bivalves and crustaceans.

9.1.2 Stock ID and possible assessment areas

There is no information available on lemon sole stock identity for the greater North Sea (including the Skagerrak and eastern English Channel areas), and the assessment is assumed to cover one-unit stock.

9.1.3 Management regulations

No specific management objectives are known to ICES. An EU TAC was set for EU waters of ICES Division 2.a and Subarea 4, which is a joint TAC together with witch flounder. However, since July 2023, a TAC is set for EU waters of ICES Division 2.a, Subarea 4 and divisions 3.a and 7.d, for lemon sole only, excluding projected third country catches (EU-UK Joint Committee, 2023).

9.2 Fisheries data

9.2.1 Officially-reported landings

Both in the North Sea and in the Skagerrak and Kattegat, lemon sole is mainly a by-catch species in the fisheries for mixed demersal stocks and for plaice. Officially-reported landings in ICES Division 7.d, Subarea 4 and Division 3.a are shown in Figures 9.2.1 to 9.2.4, and in Tables 9.2.1 to 9.2.4. The time-series of officially-reported landings is not fully complete, and a number of countries have gaps in data provision.

9.2.2 ICES estimates of landings and discards

Investigations into the then-extant data for the WKNSEA data meeting (November 2017; ICES, 2018) suggested that there would be insufficient age samples to permit an age-structured catch-based assessment, so the subsequent data calls and collations have focussed on length-based data.

Commercial catch data were raised to fleet and country level using InterCatch. The benchmark meeting (ICES, 2018) discussed whether areas should be considered separately for raising discards and length compositions, but the prevailing view was that there was no evidence of distinct stocks between areas and that therefore all areas should be treated together for raising. Initial exploration demonstrated that the final discard raising was significantly influenced by a small number of métiers with discard ratios greater than 1.5 (in other words, those métiers for which discards/landings > 1.5). Subsequently, these métiers were discounted in calculating raising factors as they were thought to be non-representative for a high-value stock such as lemon sole. Otherwise, discards for all unsampled fleets were inferred by a discard rate generated using all sampled fleets (weighted by the landings CATON), as it was not thought likely that discard rates for an (essentially) bycatch stock would vary a great deal between different métiers (apart from the extreme and unrepresentative examples discussed above).

Length-distribution allocations were conducted in the same way (weighted by mean numbers at length), with the only distinction being made between landings and discards. Length samples are reasonably well-spread across the main countries catching lemon sole, and length-based allocations are likely to be sufficiently representative.

Both BMS (Below Minimum Size) landings and logbook-recorded discards were included with discards for length-allocation purposes as the length distributions are likely to be similar. For the years 2019-2024, there were no submissions for logbook-recorded discards (0 tonnes). Scotland

provided submissions of BMS landings for 2019 (0.224 tonnes in Subarea 4), 2022 (0.04 tonnes in Subarea 4), 2023 (0.04 tonnes in Subarea 4) and 2024 (0.04 tonnes in Subarea 4), whilst England submitted data for 2020 (0.216 tonnes in Subarea 4) and 2021 (0.021 tonnes in Subarea 4, and 0.008 tonnes in Division 7.d).

There were no noted revisions to InterCatch submissions for years prior to 2021. For WGNSSK 2025, England submitted new InterCatch data for the years of 2021, 2022 and 2023, which resulted in minor differences. These years were raised again and all the calculations using InterCatch values were redone. In 2021, the total landings changed from 2104 to 2106 tonnes, while the discards increased from 501 to 668 tonnes. For 2022, there was 1 tonne difference in landings (1506, but now 1507 tonnes), while the discards increased from 345 tonnes to 454 tonnes. For 2023, there was 2 tonnes difference in landings (1471 but now 1473 tonnes), while the discards amounted from 300 to 418 tonnes.

InterCatch summary plots are given in Figures 9.2.5 to 9.2.8. The resultant estimates for landings and discards for 2002–2024, along with official landings for 1968–2024, are given in Table 9.2.5 and Figure 9.2.9. The ICES estimates for landings and discards by area are given in Table 9.2.6. The 2013 discard estimate is very high – the problem appears to originate in the discard estimates provided by the Netherlands, but it has not yet been possible to address this error. The abundances at length in the Dutch submissions are an order of magnitude higher than for any other year or country, for fish less than 210 mm. This gives rise to the high discard estimate in 2013. The issue was avoided in the F_{MSY} proxy analysis (see Section 9.6) by removing the 2013 data, but this issue has not yet been addressed for the yield analysis.

In the North Sea, eastern English Channel and Skagerrak, lemon sole was managed using a combined TAC with witch flounder up until 2023. The ICES estimates of landings (catches since 2019 implementation of a catch TAC) for lemon sole and witch are compared with the joint TAC in Figure 9.2.10, which shows that the joint TAC is underutilised with the exception of 2006, 2007, 2016, 2022 and 2023. From 2024 onwards, the TAC of lemon sole and witch are no longer combined (EU-UK Joint Committee, 2023).

9.3 Survey data series

9.3.1 Stock distributions

Figure 9.3.1a, b and c display the distribution of the abundance of lemon sole in the greater North Sea obtained from IBTS Q1 (2020–2025), IBTS Q3 data (2019–2024) and BTS Q3 data (2019–2024). Generally, the highest concentrations of lemon sole occur in the central to northern areas of the North Sea. Lemon sole abundance in the southern North Sea is generally low throughout the years.

In 2022, the IBTS Q1 and BTS Q3 survey were impacted by problems with weather, vessel reliability and Covid contamination on board, as can be seen by the reduced survey hauls for that year (Figure 9.3.1a, c). However, the remaining survey hauls were thought to have been sufficiently representative of overall lemon sole abundance and density for several reasons. The missing hauls in the BTS Q3 data were located in area 7d and along the west coast of Denmark in area 4 (Figure 9.3.1d). These missing locations are characterized by occurrence of zero hauls in previous years. Lemon sole typically shows high abundance in surveys in the central North Sea. Furthermore, the Q3 survey index is modelled using the delta GAM approach and the model includes a spatial effect and combines both Q3 BTS and Q3 IBTS survey, the latter having full coverage in 2022.

The German BTS survey used a commercial vessel instead of their usual research vessel in 2023 and this was the case for the Belgian BTS in 2024. However, the resulting index was not sensitive to the change in vessel.

9.3.2 Maturity and weights-at-age

Following the Stock Annex, maturities were assumed to be fixed through time and set to the following values by age:

Age	Prop. Mature
1	0.00
2	0.72
3 and older	1.00

Weights-at-age were also estimated following the Stock Annex procedure. The mean weights at each age and year were calculated from data in the Age Length Key by Sex and Maturity dataset of the IBTS Q1, IBTS Q3 series and BTS Q3 (ICES DATRAS). For each age, the time-series of available weights were plotted together, positioned so that Q1 weights were at $y+0.25$ and Q3 weights at $y+0.75$ (additional mean points were added at the start of each time-series to enable extrapolation). A *loess* smoother (span = 1) was then fitted through all points for each age, so that the final estimate was (effectively) a smoothed average of consecutive weight estimates. The fitted values are summarised in Figure 9.3.2 and Table 9.3.1. Small differences by age are likely the result of small changes in the weight entries in the Age Length Key by Sex and Maturity dataset related to updated weight-length keys in DATRAS. We also note that estimates for 2025 are included here: these are not currently used in the SURBAR stock assessment which concludes in 2024, but they are included for completeness.

Natural mortality (M) estimates for lemon sole are not available. For current advisory purposes, however, estimates of M are not required, as the assessment is survey-based and hence estimates total mortality Z .

9.3.3 Relative abundance indices

The GAM estimation approach (Berg *et al.*, 2014) was used by WGNSSK to generate updated Q1 (IBTS) and combined Q3 (IBTS and BTS) survey series for lemon sole. The new series are summarised in Table 9.3.2 and Figures 9.3.3 (bivariate scatterplots), 9.3.4 (catch curves), and 9.3.5 (time series by age and cohort). These summaries indicate that the ability of the survey indices (particularly Q1) to track year-class strength is very limited. For example, in Figure 9.3.3, most of the pairwise comparisons do not show significant correlations (and some comparisons are negative).

Not shown here is a significantly negative correlation between age 1 and age 2 for the Q1 (IBTS) index – this suggests that the Q1 (IBTS) age 1 index will give an incorrect impression of subsequent year-class strength, which is likely to be due to very small sample sizes at that age. The Stock Annex for this assessment calls for the full age range (1-5) to be used from the Q1 (IBTS) series. Following the presentation of the exploratory survey analyses at the 2018 meeting, WGNSSK concluded that the age-1 data from the Q1 (IBTS) survey should not be used to indicate stock trends. Therefore, the Q1 (IBTS) survey index was limited to ages 2-5 for assessment purposes at the 2019 meeting, and this has been continued during 2020-2025. Although biological

sampling for IBTS Q1 has increased in 2025 (compared to 2024), it is still low (Table 9.3.3). This is due to lemon sole not being a standard species of the survey. This issue was raised during the WGNSSK 2024 meeting and will be discussed in more detail in a dedicated lemon sole age reading workshop being held in the Spring of 2026. Survey coverage and biological sampling for the combined IBTS and BTS Q3 survey index is considered sufficient (Table 9.3.4).

It was recommended to run SURBAR including all available survey data. The resulting relative SSB for years where all surveys are available were used as index of abundance (2007-2024, excluding the year 2025 for which only the IBTS Q1 index is available) in the application of the empirical *chr* rule to give advice.

9.4 SURBAR stock assessment

- The SURBAR assessment was conducted according to the run-time settings specified in the Stock Annex, namely:
- The age- and year-effect smoother λ was set to 3.
- Mean mortality Z was calculated over ages 3-5.
- The reference age a_r for age-effect estimates was set to 3.
- GAM-estimated survey indices from both Q1 (IBTS) and Q3 (IBTS & BTS) were used.
- Catchability for ages was set as $q_1 = 0.1$, $q_2 = 0.5$ and $q = 1.0$ for all older ages. This correction is intended to reduce the impact on the analysis of the observed pronounced “hooks” at the top of the survey catch curves for this stock (see Figure 9.3.4). A proposal for a systematic method of determining catchability corrections to straighten catch curves prior to SURBAR assessment was presented at the WGNSSK 2020 meeting. While promising, this method remains in development and will be revisited in a future WGNSSK meeting.
- No downweighting of ages in the SURBAR SSQ estimation was used.

The SURBAR stock summary is given in Table 9.4.1, and the corresponding output plots are given in Figures 9.4.1 to 9.4.4. The stock summary (Figure 9.4.1) shows that mean Z_{3-5} has remained relatively constant since 2009, although values are very low and the confidence intervals overlap $Z = 0$ for most years. The catch curves for the surveys (Figure 9.3.4) are domed and very shallow, and remain shallow even when the catchability revision is applied, so SURBAR indicates very low mean Z_{3-5} . Both SSB and TSB are estimated with more certainty than mean Z_{3-5} , and both show steady declines since 2016. Finally, recruitment at age 1 has fluctuated without trend for much of the time series, with indications of steady increase since 2020 (although the uncertainty about the recent estimates are relatively large). The latest recruitment estimate was lower than last year. However, uncertainty is estimated to be larger for the most recent values.

Log survey residuals (Figures 9.4.2) show that the Q3 index fits the SURBAR model better than the Q1 index, with lower residuals (in general) and less trends through time. Consequently, the assessment is driven more directly by the Q3 index – this is to be expected given the problems with the Q1 index highlighted in Section 9.3.3 above. There are three outliers in the Q3 index (age 1 in 2013, 2015 and 2017, age 2 in 2013), but sensitivity runs reducing the SSQ estimation weighting on these points suggested that their influence on likely advice was not significant (ICES, 2018). The parameter estimates are summarised in Figure 9.4.3.

The retrospective analysis in Figure 9.4.4 shows little retrospective bias or noise for any of the key SURBAR outputs. This is a simple retrospective peel, and does not account for any retrospective bias or noise in the delta-GAM index estimation or the calculation of stock weights-at-age. The survey indices from five subsequent WGNSSK meetings are plotted in Figure 9.4.5, which shows good consistency from year to year with the exception of a small step change five years ago. Retrospective patterns in the stock weights-at-age have not been explored, but the

overall consistency in survey indices suggests that a true retrospective analysis for lemon sole would be similar to the simple retrospective analysis presented in Figure 9.4.4.

Finally, the run presented here assumes a lambda smoother of 3.0. A low lambda setting ($\lambda = 1.0$) results in large interannual variations in all outputs, driven by survey noise and the difficulty in following cohorts. Increasing the lambda smoother leads to less variation, as expected, and the outputs for $\lambda = 3.0$ and $\lambda = 5.0$ are very similar, increasing confidence that the setting $\lambda = 3.0$ is probably reasonable (increasing lambda further doesn't lead to much change). Further methodological work on systematically defining the appropriate lambda smoother for a given assessment is underway, and will be presented at a future WGNSSK meeting.

9.5 Generating advice for the data-limited lemon sole stock

North Sea lemon sole is currently managed according to the following advice, given in June 2024:

ICES advises that when the MSY approach is applied, catches in 2025 should be no more than 1450 tonnes.

Following the release of the 2019 advice, ICES has been requested to issue annual advice for North Sea lemon sole.

Advice is given according to the new WKLIFE DLS approach following the *chr* rule (ICES 2022, Figure 9.5.1). This is provided for lemon sole in Section 9.5.2, but first in Section 9.5.1 we consider length-based F_{MSY} proxy estimation (as this is required for the WKLIFE approach).

9.5.1 Length-based F_{MSY} proxy estimation

Length-based indicators (LBIs) for F_{MSY} proxies were estimated for North Sea lemon sole, following the standard approach outlined by WKLIFE (ICES, 2017) and WKPROXY (ICES, 2016), and stipulated in the lemon sole Stock Annex by the 2018 benchmark meeting (ICES, 2018). Data were taken from the length samples submitted to InterCatch for 2002–2024.

The original InterCatch length distributions are given in Figure 9.5.2, from which erroneous length submissions for fish less than 200 mm in 2013 can clearly be seen. These seem to arise from Dutch discard samples, which could not be corrected prior to the WGNSSK meeting (see also Section 9.2.2). To address this without correcting the input data, the 2013 data were removed from the analysis (this has no impact on the final conclusions). Figure 9.5.3 shows the result of this, along with the removal of all fish less than 100 mm (to prevent the misspecification of length at first capture L_c). Finally, the widths of the length bins were doubled to produce smoother distributions for the calculation of L_c (Figure 9.5.4).

Figure 9.5.5 shows a logit maturity ogive fitted to maturity data from the Q1 (IBTS) and Q3 (IBTS & BTS) survey records, using a binomial GLM with a logit link. This analysis indicates that a suitable estimate of L_{mat} would be 136 mm.

Figure 9.5.6 shows an estimated L_∞ value of 281 mm and associated K value of 0.383/yr, derived from all available survey data. In order to make sure the K value was indeed a correct assumption for the growth rate parameter of lemon sole, an investigation was performed for females and males separately, using all available survey data (Fig. 9.5.6.1). The next step was to check, if, when t_0 was not set to zero, we would obtain different values for K (Fig. 9.5.6.2). In any scenario the K value remained high and within the range of 0.32 and 0.45, so the value of the multiplier m would still remain at 0.5.

Assessment year	K estimate
2024	0.383
2025	0.383

WGNSSK was concerned that the survey-derived value of 281 mm was likely to be too low, given the possibility (although uncertain) that survey catchability for older fish may be poor. Two alternative estimates of L_{∞} were hence considered – the longest fish observed in the commercial fishery landings data (695 mm), and a trimmed alternative based on the 99 percentile of the commercial catch length distribution (385 mm, collated over all available years). The estimates are summarised in Figure 9.5.7. Given L_{max} , WGNSSK proposed that L_{∞} should be derived from the following equation (García-Carreras *et al.*, 2016):

$$\log_{10} L_{\infty} = 0.068260 + 0.969112 \log_{10} L_{max}$$

The resultant estimates are then:

Basis	L_{max}	L_{∞}
Trimmed L_{max}	385 mm	375 mm
Observed L_{max}	695 mm	664 mm
Survey data	-	281 mm

WGNSSK conclude that L_{∞} should be set to 375 mm (as for last year, and highlighted above), as the estimate of 664 mm does not seem to be representative of the bulk of the stock, and the survey-based estimate (281 mm) may be biased downwards by reduced catchability for older lemon sole in the surveys.

This estimate of L_{∞} , along with the new estimate of L_{mat} were then used in an LBI estimation run which is summarised in Figures 9.5.8 and 9.5.9, and Table 9.5.1. The key points are:

- Length at first catch (L_c) is above L_{mat} for most of the time-series, which indicates fewer immature individuals in the catches.
- The ratio of the mean length of the upper 5th percentile of catches to L_{∞} is around 1.0 throughout the time series, which would suggest a reasonable number of large (and hence old) fish in the population. The fitted age-length curve in Figure 9.5.6 would indicate that it is reasonable to assume a large lemon sole is older than a small one. There is a spread, but the underlying trend seems well-defined.
- The L_{mean}/L_{opt} ratio is greater than 1.0 for most of the time series, which suggests that exploitation is targeting the most productive length classes, where $L_{opt} = 2/3L_{\infty}$.
- $L_{mean}/L_{F=M}$ is greater than 1.0 for all years in the time-series, which indicates that this stock is being fished at a rate less than (or close to) F_{MSY} . $L_{F=M}$ is calculated as $L_{F=M} = 0.75L_c + 0.25L_{\infty}$.

The LBI results suggest that immature fish are well protected, and that the catch length distribution is not truncated at larger sizes: under optimal and sustainable exploitation the mean length in the catch is expected to be higher than the value observed, and this is the case here. The fact that the ratio of $L_{mean}/L_{F=M}$ is greater than 1.0 throughout the time-series would suggest that F_{MSY} is **not** being exceeded for this stock.

9.5.2 Application of the WKLIFE X DLS approach

The WKLIFE DLS decision tree (ICES, 2020) has been replicated for convenience in Figure 9.5.1. Following this through for lemon sole leads to the following conclusions:

1. There is no accepted SPiCT assessment for lemon sole.
2. Indices of abundance, commercial catch length data, and an estimate of the von Bertalanffy K parameter are all available.
3. For lemon sole, $K = 0.383\text{yr}^{-1}$.
4. Hence the *chr* rule (method 2.2) should be used to provide advice, given that $0.32\text{yr}^{-1} \leq K \leq 0.45\text{yr}^{-1}$.

ICES (2020) notes that the *chr* rule should be based on an index of biomass. In the case of lemon sole, this could be generated by the existing SURBAR analysis. WGNSSK 2022 used an empirical biomass index based on the Q1 survey considering that it was important to use the most recent data available offered by the IBTS Q1 empirical index. However, in 2023, due to low age sampling for the IBTS Q1 survey index (section 9.3.3), WGNSSK concluded that using the SSB from the SURBAR assessment from years which include all surveys is preferable for use as index of biomass in the *chr* rule to provide the advice. The same approach regarding the SSB (from the SURBAR assessment) was used in WGNSSK 2025.

For comparison, the empirical survey indices (see Table 9.3.2) are calculated by:

$$I_y = \sum_{a=2}^5 i_{a,y} W_{a,y} m_{a,y}$$

where i is the survey index, W is the stock weight, and m is the proportion mature. Ages 2-5 only are used, because those are the ages for which there were sufficient observations in the survey catch data (> 50 samples per year was used as a threshold).

The *chr* rule states that:

$$C_{y+1} = I_y HR_{MSY\ proxy} b m$$

The components of this formula were estimated as follows.

- I_y is the biomass index for year y . In this case, using the SURBAR SSB index until 2024 (Figure 9.4.1; Table 9.4.1), $I_y = \mathbf{0.629}$.
- $HR_{MSY\ proxy}$ is the average of the ratio of catch C to the biomass index I , from SURBAR SSB estimates calculated in WGNSSK 2025, calculated across all years for which $L_{mean}/L_{F=M} > 1$. The comparison between L_{mean} and $L_{F=M}$ is shown in Figure 9.5.10, from which it can be seen that all years could be considered (excluding 2013). However, for this year's assessment the same year range should be used in the calculation of $HR_{MSY\ proxy}$ as last year when the method was first applied to this stock: 2007 to 2022. The ratio C/I is shown in Figure 9.5.11 and Table 9.5.2, and the average is **4003**.
- $b = \min\{1, I_y/I_{trigger}\}$. The value used for $I_{trigger}$ would often be $1.4I_{loss}$, where I_{loss} is the lowest observed biomass index value ($I_{loss} = 0.715$). However, for a stock such as lemon sole for which historical exploitation has been relatively light (as shown by the LBI analysis), it is appropriate to set $I_{trigger} = I_{loss}$.
- m is a multiplier intended to avoid biomass declining below B_{lim} . Lemon sole has a relatively fast observed growth rate, and in this situation the WKLIFE decision tree recommends that $m = \mathbf{0.5}$.

Using these estimates in the formula gives:

$$C_{y+1} = 0.629 \times 4002.867 \times 0.88 \times 0.5 = 1107 \text{ t.}$$

This represents a reduction in the advice of 23.6%, and as $b < 1$, the stability clause should not be applied. This results in an advice of **1107 t**.

The change in advice (−23.6%) is due to a decreasing biomass index.

9.6 Conclusions and further work

Median values of SURBAR estimates for SSB indicate a declining trend since 2016 which is reflected in the advice. The recruitment estimates suggest that recruitment may be increasing in recent years, although retrospective noise problems indicate that this should be treated as being very uncertain.

The estimation of status relative to F_{MSY} proxies indicates that fishing is occurring at or below F_{MSY} , which was also the conclusion in the WGNSSK meetings in 2017–2025.

These conclusions are based on stock dynamics indicated by a survey index and a survey-based assessment, and the inability (in many cases) of the available surveys to track year-class strength is a weak point of the advice. An important issue for the development of new advice would be to reconsider the survey series used – further work may indicate an alternative method of collating the survey data that could be more appropriate for lemon sole.

9.7 Issues list

9.7.1 Data and assessment

The current survey indices used for North Sea lemon sole are not able to track cohort strength on a consistent basis, and they exhibit generally poor catchability characteristics which limit the reliability of the advice based thereon. It would be very beneficial to be able to include commercial catch data in the assessment in order to improve reliability and reduce variability. Unfortunately, age data are lacking from commercial catch data, so a (spatial) length-based assessment using both catch and survey data should be explored (for example, Stock Synthesis 3). It may also be appropriate to reconsider the use of SPiCT as the basis for advice – this was tested and shown not to work at the last benchmark (ICES, 2018), but new survey series may improve the applicability of the method.

Natural mortality is assumed to be time-invariant in the current assessment. The potential of using key MSVPA runs to provide time-varying natural mortality estimates for North Sea lemon sole should be explored. Similarly, maturity is assumed to be time-invariant, and this needs to be considered again. In June 2024, a workshop on the maturity staging of lemon sole (WKMSLEM) was held in Belgium. Below is a list of the conclusions from this workshop:

- Maturity stage descriptions: a detailed description of the maturity stages for females and males lemon sole gonads were created.
- Lemon sole maturity development: a deeper understanding about the progress of maturity stages within a season was undertaken for both males and females.
- Reliability of macroscopic maturity staging. In this report, it is stated that: “WKSLEM recommends that WGBIOP investigates the possibility for a maturity coding that relates to the activity in the spawning season (A vs. ‘not A’ for B-C-D) would increase the reliability of the maturity stage recordings, and so, improve the information for maturity ogives.”
- Sampling scheme: because lemon sole has a long maturity season, WKMSLEM recommended that, whenever sampling occurs, the maturity stage should also be analysed.

More detailed information about this workshop can be found in the meeting report (ICES, 2024)

9.7.2 Forecast

Lemon sole advice is currently based on the WKLIFE DLS approach. If a more quantitative assessment can be generated, then there may be a requirement (and opportunity) to develop a forecast methodology, and this will need to be addressed when appropriate.

9.8 References

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9.9 Tables and figures

Table 9.2.1. Lemon sole in Subarea 4 and divisions 3.a and 7.d. Official lemon sole landings by area (tonnes). *Preliminary.

Official landings									
Year	3.a	4	7.d	Total	Year	3.a	4	7.d	Total
1950	307	3754	208	4269	1986	639	5047	251	5937
1951	248	4710	314	5272	1987	669	5516	310	6495
1952	243	4922	298	5463	1988	642	5898	258	6798
1953	132	5440	386	5958	1989	693	5967	364	7024
1954	128	3972	534	4634	1990	872	6190	423	7485
1955	102	3836	141	4079	1991	734	6618	428	7780
1956	96	3395	103	3594	1992	952	6126	364	7442
1957	78	3419	102	3599	1993	1156	5839	422	7417
1958	94	3104	82	3280	1994	803	5262	695	6760
1959	130	3647	82	3859	1995	714	4712	877	6303
1960	153	4035	66	4254	1996	635	4737	1151	6523
1961	161	4900	108	5169	1997	768	4727	563	6058
1962	93	4630	101	4824	1998	868	6466	346	7680
1963	99	3791	66	3956	1999	844	6316	140	7300
1964	134	4121	77	4332	2000	803	5980	388	7171
1965	164	4949	105	5218	2001	584	5389	483	6456
1966	159	5415	201	5775	2002	522	3827	474	4823
1967	191	6188	331	6710	2003	543	3688	491	4722
1968	185	6270	337	6792	2004	607	3543	424	4574
1969	215	4470	315	5000	2005	674	3444	350	4468
1970	169	3434	256	3859	2006	417	3627	246	4290
1971	173	3967	357	4497	2007	432	3892	164	4488
1972	168	3672	475	4315	2008	276	3465	234	3975
1973	214	4568	451	5233	2009	262	2690	441	3394
1974	183	4227	351	4761	2010	350	2626	224	3200
1975	317	5029	33	5379	2011	254	3369	422	4045

Official landings									
Year	3.a	4	7.d	Total	Year	3.a	4	7.d	Total
1976	361	4830	42	5233	2012	482	3123	466	4071
1977	627	5661	37	6325	2013	290	2981	492	3763
1978	705	6108	141	6954	2014	315	3019	357	3692
1979	833	6428	260	7521	2015	295	2925	260	3480
1980	722	6424	152	7298	2016	300	3268	239	3807
1981	793	5933	290	7016	2017	340	2845	158	3343
1982	735	7168	584	8487	2018	280	2641	99	3021
1983	759	8257	491	9507	2019	329	2808	104	3241
1984	595	6930	586	8111	2020	339	2221	96	2656
1985	793	6435	347	7575	2021	256	1747	89	2092
					2022	307	1135	72	1513
					2023*	293	1144	43	1481
					2024*	209	943	34	1186

Table 9.2.2. Lemon sole in Subarea 4 and divisions 3.a and 7.d. Official lemon sole landings in Division 7.d by country. *Preliminary.

Year	BEL	DNK	FRA	NED	UK	Other	Total	Year	BEL	DNK	FRA	NED	UK	Other	Total
1950	10	0	174	0	24	0	208	1986	77	0	133	0	41	0	251
1951	5	0	262	0	47	0	314	1987	81	0	185	0	44	0	310
1952	10	0	188	0	100	0	298	1988	74	0	155	0	29	0	258
1953	7	0	196	0	183	0	386	1989	68	0	252	0	44	0	364
1954	9	0	361	0	164	0	534	1990	68	0	272	0	83	0	423
1955	9	0	0	0	132	0	141	1991	83	0	272	0	73	0	428
1956	4	0	0	0	99	0	103	1992	66	0	176	0	122	0	364
1957	7	0	0	0	95	0	102	1993	36	0	311	0	75	0	422
1958	1	0	0	0	81	0	82	1994	97	0	505	0	93	0	695
1959	2	0	0	0	80	0	82	1995	138	0	584	0	155	0	877
1960	4	0	0	0	62	0	66	1996	213	0	720	0	218	0	1151
1961	1	0	0	0	106	1	108	1997	143	0	305	0	115	0	563
1962	2	0	0	0	99	0	101	1998	53	0	198	0	95	0	346
1963	3	0	0	0	63	0	66	1999	50	0	0	0	90	0	140
1964	5	0	0	0	72	0	77	2000	62	0	200	0	126	0	388
1965	16	0	0	0	89	0	105	2001	104	0	191	0	188	0	483
1966	7	0	0	0	194	0	201	2002	101	0	256	0	117	0	474

Year	BEL	DNK	FRA	NED	UK	Other	Total	Year	BEL	DNK	FRA	NED	UK	Other	Total
1967	6	0	0	0	325	0	331	2003	128	0	251	0	112	0	491
1968	8	0	0	0	329	0	337	2004	120	0	198	1	105	0	424
1969	12	0	0	0	303	0	315	2005	90	0	187	2	71	0	350
1970	16	0	0	0	240	0	256	2006	98	0	100	0	48	0	246
1971	22	0	0	0	335	0	357	2007	70	0	72	1	21	0	164
1972	18	0	0	0	457	0	475	2008	140	0	46	3	45	0	234
1973	25	0	0	0	426	0	451	2009	149	0	176	9	107	0	441
1974	16	0	0	1	334	0	351	2010	101	0	85	5	32	0	224
1975	19	0	0	0	14	0	33	2011	153	0	197	15	57	0	422
1976	24	0	0	0	18	0	42	2012	171	0	167	20	108	0	466
1977	21	1	0	0	15	0	37	2013	175	0	181	26	110	0	492
1978	45	2	63	0	31	0	141	2014	163	0	108	14	72	0	357
1979	60	0	165	0	35	0	260	2015	122	0	85	10	42	0	260
1980	33	0	109	0	10	0	152	2016	116	0	69	9	47	0	239
1981	66	0	212	0	12	0	290	2017	87	0	34	8	30	0	158
1982	96	0	406	1	81	0	584	2018	57	0	21	5	16	0	99
1983	108	0	298	0	85	0	491	2019	49	0	27	6	23	0	104
1984	110	0	367	0	109	0	586	2020	46	0	25	6	19	0	96

Year	BEL	DNK	FRA	NED	UK	Other	Total	Year	BEL	DNK	FRA	NED	UK	Other	Total
1985	117	0	164	0	66	0	347	2021	38	0	18	7	26	0	89
								2022	39	0	17	4	11	0	72
								2023*	22	0	11	3	6	0	43
								2024*	20	0	7	2	5	0	34

Table 9.2.3. Lemon sole in Subarea 4 and divisions 3.a and 7.d. Official lemon sole landings in ICES Subarea 4 by country. *Preliminary.

Year	BEL	DNK	FRA	GER	NED	NOR	UK	Other	Total	Year	BEL	DNK	FRA	GER	NED	NOR	UK	Other	Total
1950	112	435	139	31	156	0	2855	26	3754	1973	566	478	0	73	217	16	3218	0	4568
1951	115	845	90	21	167	0	3430	42	4710	1974	486	447	0	59	269	0	2966	0	4227
1952	98	391	227	26	168	0	3953	59	4922	1975	748	521	0	83	299	0	3367	11	5029
1953	73	409	189	18	132	0	4590	29	5440	1976	493	506	0	68	308	0	3443	12	4830
1954	2	272	177	24	112	0	3368	17	3972	1977	618	321	0	71	262	0	4387	2	5661
1955	49	311	0	15	78	0	3374	9	3836	1978	760	517	28	54	231	0	4518	0	6108
1956	48	222	0	19	58	0	3034	14	3395	1979	674	876	136	41	390	0	4308	3	6428
1957	39	249	0	24	64	0	3032	11	3419	1980	484	599	102	49	303	0	4885	2	6424
1958	30	171	0	13	43	0	2835	12	3104	1981	555	605	237	39	412	0	4084	1	5933
1959	85	242	0	40	43	0	3226	11	3647	1982	879	670	419	52	759	0	4386	3	7168
1960	155	577	0	46	67	0	3178	12	4035	1983	1122	735	402	28	1009	0	4957	4	8257
1961	286	488	0	79	102	0	3934	11	4900	1984	1144	567	344	22	0	0	4850	3	6930
1962	175	501	0	54	106	0	3794	0	4630	1985	989	555	157	26	0	0	4703	5	6435
1963	365	222	0	36	71	0	3097	0	3791	1986	511	577	103	16	0	0	3839	1	5047
1964	484	358	0	62	75	0	3142	0	4121	1987	448	742	174	14	0	0	4137	1	5516
1965	562	385	0	91	93	0	3818	0	4949	1988	539	639	184	14	301	0	4220	1	5898
1966	594	548	0	98	65	0	4110	0	5415	1989	441	828	176	40	397	0	4083	2	5967
1967	601	791	0	136	61	0	4599	0	6188	1990	491	1007	208	49	0	0	4431	4	6190
1968	422	775	0	96	34	0	4943	0	6270	1991	544	1099	250	41	0	12	4666	6	6618
1969	292	639	0	80	36	0	3423	0	4470	1992	577	1149	177	30	0	13	4175	5	6126
1970	241	307	0	52	58	0	2776	0	3434	1993	525	966	240	37	0	9	4059	3	5839
1971	348	514	0	54	122	0	2929	0	3967	1994	436	597	436	27	0	11	3754	1	5262
1972	423	530	0	59	130	0	2530	0	3672	1995	588	585	412	70	0	9	3046	2	4712

Year	BEL	DNK	FRA	GER	NED	NOR	UK	Other	Total	Year	BEL	DNK	FRA	GER	NED	NOR	UK	Other	Total
1996	592	547	534	67	0	18	2976	3	4737	2009	389	897	88	63	294	31	927	2	2690
1997	504	499	224	76	0	29	3391	4	4727	2010	376	821	32	102	323	36	935	2	2626
1998	815	796	197	149	838	23	3643	5	6466	2011	387	999	60	96	641	27	1157	2	3369
1999	662	1015	0	62	681	24	3866	6	6316	2012	406	999	34	61	587	30	1003	2	3123
2000	711	1277	184	72	492	17	3222	5	5980	2013	527	649	28	67	479	16	1214	2	2981
2001	694	1281	191	77	451	22	2666	7	5389	2014	650	626	27	63	424	23	1203	3	3019
2002	604	971	190	116	402	17	1521	6	3827	2015	443	794	17	82	453	12	1120	3	2925
2003	517	1008	239	136	369	16	1399	4	3688	2016	448	1055	15	82	443	23	1197	5	3268
2004	667	1113	120	81	355	12	1192	3	3543	2017	345	1032	4	43	360	14	1042	4	2845
2005	595	1057	102	85	402	13	1188	2	3444	2018	370	815	9	52	349	14	1029	3	2641
2006	552	968	57	183	412	13	1440	2	3627	2019	467	671	8	47	475	13	1122	4	2808
2007	542	1136	66	143	367	23	1609	6	3892	2020	376	497	9	32	385	5	912	6	2221
2008	527	925	48	119	434	26	1383	3	3465	2021	312	331	2	33	422	7	637	3	1747
										2022	150	225	1	29	244	7	475	3	1135
										2023*	262	167	2	29	98	14	569	3	1144
										2024*	202	122	1	17	90	10	496	4	942

Table 9.2.4. Lemon sole in Subarea 4 and divisions 3.a and 7.d. Official landings in Division 3.a by country. *Preliminary.

Year	BEL	DNK	GER	NED	SWE	Other	Total	Year	BEL	DNK	GER	NED	SWE	Other	Total
1950	0	100	1	0	206	0	307	1986	7	576	0	0	56	0	639
1951	0	74	1	0	173	0	248	1987	24	577	0	0	68	0	669
1952	0	64	0	0	179	0	243	1988	11	569	0	6	56	0	642
1953	0	35	0	0	97	0	132	1989	8	610	0	0	75	0	693
1954	0	33	0	0	95	0	128	1990	16	782	0	0	74	0	872
1955	0	29	0	0	73	0	102	1991	11	640	0	0	83	0	734
1956	0	33	0	0	63	0	96	1992	22	793	0	0	120	17	952
1957	0	27	0	0	51	0	78	1993	14	980	4	0	141	17	1156
1958	0	38	0	0	56	0	94	1994	10	648	2	0	127	16	803
1959	0	71	0	0	59	0	130	1995	27	576	2	0	91	18	714
1960	0	95	1	0	57	0	153	1996	0	513	1	0	97	24	635
1961	0	90	0	0	71	0	161	1997	0	628	2	0	115	23	768
1962	0	92	1	0	0	0	93	1998	0	743	3	0	100	22	868
1963	0	99	0	0	0	0	99	1999	0	731	3	0	88	22	844
1964	0	133	1	0	0	0	134	2000	0	722	1	0	65	15	803
1965	0	163	1	0	0	0	164	2001	0	511	1	0	53	19	584
1966	0	159	0	0	0	0	159	2002	0	457	4	0	41	20	522
1967	0	189	1	0	0	1	191	2003	0	451	6	30	35	21	543
1968	0	184	0	0	0	1	185	2004	0	472	5	82	29	19	607
1969	0	215	0	0	0	0	215	2005	0	468	5	147	38	16	674
1970	0	169	0	0	0	0	169	2006	0	321	8	40	32	16	417
1971	0	173	0	0	0	0	173	2007	0	374	5	16	18	19	432
1972	0	168	0	0	0	0	168	2008	0	239	7	3	15	12	276
1973	0	214	0	0	0	0	214	2009	0	233	4	1	15	9	262
1974	0	183	0	0	0	0	183	2010	0	286	3	35	19	8	350
1975	0	263	1	1	52	0	317	2011	0	223	3	0	12	16	254
1976	10	294	1	19	37	0	361	2012	0	446	3	0	15	18	482
1977	9	528	2	37	51	0	627	2013	0	259	3	5	10	12	290

Year	BEL	DNK	GER	NED	SWE	Other	Total	Year	BEL	DNK	GER	NED	SWE	Other	Total
1978	4	628	2	12	59	0	705	2014	0	276	7	12	14	6	315
1979	7	704	1	10	111	0	833	2015	0	250	4	25	10	6	295
1980	12	622	0	0	87	1	722	2016	0	265	5	16	8	6	300
1981	1	710	0	3	75	4	793	2017	0	314	1	11	7	7	340
1982	2	647	0	9	77	0	735	2018	0	253	5	14	6	2	280
1983	3	636	0	10	110	0	759	2019	0	292	1	30	5	1	329
1984	6	525	0	0	64	0	595	2020	0	286	3	45	4	1	339
1985	0	729	0	0	64	0	793	2021	0	226	1	25	3	1	256
								2022	0	281	1	20	4	1	307
								2023*	0	283	1	5	3	1	293
								2024*	0	194	1	9	3	2	209

Table 9.2.5. Lemon sole in Subarea 4 and divisions 3.a and 7.d. Official landings, ICES estimates of landings and discards for areas 3.a, 4 and 7.d. *Preliminary.

Year	Official landings	ICES Landings	ICES Discards	ICES Total Catch	Discard rate
1950	4269				
1951	5272				
1952	5463				
1953	5958				
1954	4634				
1955	4079				
1956	3594				
1957	3599				
1958	3280				
1959	3859				
1960	4254				
1961	5169				
1962	4824				
1963	3956				
1964	4332				

Year	Official landings	ICES Landings	ICES Discards	ICES Total Catch	Discard rate
1965	5218				
1966	5775				
1967	6710				
1968	6792				
1969	5000				
1970	3859				
1971	4497				
1972	4315				
1973	5233				
1974	4761				
1975	5379				
1976	5233				
1977	6325				
1978	6954				
1979	7521				
1980	7298				
1981	7016				
1982	8487				
1983	9507				
1984	8111				
1985	7575				
1986	5937				
1987	6495				
1988	6798				
1989	7024				
1990	7485				
1991	7780				
1992	7442				
1993	7417				

Year	Official landings	ICES Landings	ICES Discards	ICES Total Catch	Discard rate
1994	6760				
1995	6303				
1996	6523				
1997	6058				
1998	7680				
1999	7300				
2000	7171				
2001	6456				
2002	4823	4011	511	4522	11.30%
2003	4722	4575	1037	5612	18.48%
2004	4574	4394	635	5029	12.63%
2005	4468	4429	527	4956	10.63%
2006	4290	4294	1515	5809	26.08%
2007	4488	4468	451	4919	9.17%
2008	3975	4153	898	5051	17.78%
2009	3394	3405	996	4401	22.63%
2010	3200	3234	673	3907	17.23%
2011	4045	4030	1024	5054	20.26%
2012	4071	4099	2461	6560	37.52%
2013	3763	3725	5938	9663	61.45%
2014	3692	3645	1690	5335	31.68%
2015	3480	3480	1636	5116	31.98%
2016	3807	3834	1167	5001	23.34%
2017	3343	3315	651	3966	16.41%
2018	3021	3046	331	3377	9.80%
2019	3241	3273	605	3878	15.60%
2020	2656	2653	391	3044	12.84%
2021	2104	2107	668	2775	24.07%
2022	1513	1507	454	1961	23.13%

Year	Official landings	ICES Landings	ICES Discards	ICES Total Catch	Discard rate
2023	1481*	1473	418	1891	22.10%
2024	1186*	1180	264	1444	18.26%

Table 9.2.6. Lemon sole in Subarea 4 and divisions 3.a and 7.d. ICES estimates of landings and discards by area 3.a, 4 and 7.d (in tonnes).

Year	ICES estimates of landings				ICES estimates of discards				ICES estimate of catches
	4	3a	7d	Total landings	4	3a	7d	Total discards	Total
2002	3119	528	364	4011	299	176	37	511	4522
2003	3525	549	501	4575	878	106	53	1037	5612
2004	3366	618	410	4394	479	94	61	635	5029
2005	3426	668	335	4429	374	122	31	527	4956
2006	3624	432	238	4294	1367	122	26	1515	5809
2007	3872	437	159	4468	363	67	21	451	4919
2008	3519	290	344	4153	755	47	96	898	5051
2009	2688	270	447	3405	877	50	70	996	4401
2010	2653	361	221	3234	519	111	42	673	3907
2011	3349	264	417	4030	853	71	100	1024	5054
2012	3118	504	477	4099	2197	95	169	2461	6560
2013	2 870	298	557	3725	1631	4208	99	5938	9663
2014	2 892	322	431	3645	1442	99	149	1690	5335
2015	2 849	302	329	3480	1468	65	104	1636	5116
2016	3 251	302	281	3834	1049	57	61	1167	5001
2017	2 794	345	176	3315	571	55	25	651	3966
2018	2 643	288	114	3046	229	76	27	331	3377
2019	2 783	339	151	3273	487	74	44	605	3878
2020	2 200	345	107	2653	253	102	36	391	3044
2021	1 754	257	96	2107	564	65	39	668	2775
2022	1 125	308	74	1507	349	92	12	454	1961
2023	1137	290	46	1473	356	37	25	418	1891

Year	ICES estimates of landings				ICES estimates of discards				ICES estimate of catches
	4	3a	7d	Total landings	4	3a	7d	Total discards	Total
2024	931	207	42	1180	215	30	19	264	1444

Table 9.3.1. Lemon sole in Subarea 4 and divisions 3.a and 7.d. Estimates of mean weight-at-age from 2005-2025

Year	Age 1	Age 2	Age 3	Age 4	Age 5	Age 6	Age 7	Age 8	Age 9
2005	0.0871	0.0763	0.1209	0.2261	0.3058	0.3398	0.3760	0.2070	0.2504
2006	0.0774	0.0756	0.1225	0.2257	0.3072	0.3356	0.3672	0.2325	0.2575
2007	0.0685	0.0745	0.1234	0.2244	0.3071	0.3319	0.3602	0.2572	0.2663
2008	0.0603	0.0731	0.1235	0.2220	0.3054	0.3289	0.3551	0.2807	0.2758
2009	0.0528	0.0713	0.1229	0.2186	0.3022	0.3264	0.3515	0.3027	0.2857
2010	0.0460	0.0694	0.1218	0.2144	0.2979	0.3240	0.3495	0.3237	0.2967
2011	0.0398	0.0670	0.1199	0.2094	0.2918	0.3226	0.3495	0.3423	0.3066
2012	0.0342	0.0645	0.1176	0.2031	0.2845	0.3215	0.3502	0.3606	0.3193
2013	0.0293	0.0615	0.1145	0.1961	0.2765	0.3201	0.3564	0.3792	0.3326
2014	0.0258	0.0578	0.1102	0.1887	0.2663	0.3225	0.3649	0.3901	0.3413
2015	0.0226	0.0545	0.1063	0.1809	0.2552	0.3224	0.3687	0.3972	0.3507
2016	0.0199	0.0514	0.1025	0.1724	0.2445	0.3203	0.3706	0.4005	0.3584
2017	0.0176	0.0484	0.0986	0.1642	0.2328	0.3156	0.3677	0.3997	0.3569
2018	0.0151	0.0460	0.0948	0.1556	0.2222	0.3060	0.3593	0.3944	0.3489
2019	0.0128	0.0439	0.0912	0.1475	0.2118	0.2939	0.3497	0.3841	0.3365
2020	0.0107	0.0420	0.0875	0.1396	0.2018	0.2796	0.3391	0.3688	0.3194
2021	0.0087	0.0405	0.0840	0.1319	0.1927	0.2634	0.3265	0.3497	0.2972
2022	0.0069	0.0392	0.0806	0.1246	0.1839	0.2449	0.3119	0.3281	0.2701
2023	0.0054	0.0383	0.0772	0.1174	0.1756	0.2241	0.2951	0.3028	0.2383
2024	0.0040	0.0376	0.0738	0.1104	0.1676	0.2006	0.2759	0.2740	0.2017
2025	0.0028	0.0373	0.0705	0.1037	0.1602	0.1743	0.2542	0.2420	0.1600

Table 9.3.2. Lemon sole in Subarea 4 and divisions 3.a and 7.d. GAM-estimated survey indices for Q1 (upper: NS IBTS) and Q3 (lower: NS IBTS + BTS).

NS Lemon Sole: IBTS Q1; Last group is NOT a plus-group. Calculations made on 16/04/2025 at 10:31:08.				
2007	2025			
1	1	0.091069	0.091069	
2	5			
1	118.495	529.1118	456.9612	968.09
1	350.8569	515.9129	265.2165	267.4536
1	337.5443	276.9217	285.3036	119.4183
1	425.6023	709.1256	966.8698	244.0228
1	459.3928	1065.247	636.123	557.4668
1	1559.552	2084.872	721.4739	332.4767
1	531.9133	820.1106	959.5977	382.7468
1	642.1975	1351.63	953.6286	211.1628
1	345.9427	1760.64	1176.963	354.1158
1	852.0072	1687.37	1009.85	410.3663
1	581.4378	1046.82	1106.171	405.185
1	419.6703	749.5825	318.4611	299.704
1	638.534	1564.242	859.8885	261.0318
1	693.1543	1358.099	722.2843	206.5343
1	340.8155	849.0753	1252.08	190.3517
1	848.2421	1639.332	901.3343	578.6723
1	1330.057	1506.448	555.3557	111.9227
1	97.1028	562.4328	956.5203	684.2473
1	431.0549	1351.465	1467.261	446.068

NS Lemon Sole: BTS & IBTS Q3; Last group is NOT a plus-group. Calculations made on 16/04/2025 at 11:33:00									
2005	2024								
1	1	0.6222967	0.6222967						
1	9								
1	195.3639	1669.6444	1793.5393	1604.1839	853.7579	1243.6217	505.7881	343.8957	527.1841
1	128.2349	1086.3724	1854.7346	1471.0402	1514.0394	800.3961	1201.5084	422.3294	204.4199
1	706.281	1693.0115	2176.628	1748.8239	1491.9906	1521.8358	657.6959	813.0578	361.7427
1	248.1691	2195.2341	2264.3701	1640.8395	1643.5725	749.535	1078.1822	430.0682	399.3935
1	575.4412	1565.6804	2582.7362	1544.2915	1044.4426	1039.4164	317.8695	682.7342	86.7446
1	518.4026	1199.4344	1934.6992	1617.6438	2034.2246	1463.8161	1375.9757	563.6637	626.3917
1	178.9359	3062.0364	3523.9675	1977.1672	2399.5409	1862.8136	852.8743	1253.9625	356.0894
1	432.4498	2404.9118	3284.5109	2459.0022	1730.6378	1315.6607	970.5506	704.3895	932.0347
1	20.8181	355.9301	2039.0629	3379.0062	2194.9097	2102.4028	1805.5272	1200.6652	466.6724
1	427.7237	1028.318	2519.0979	3242.5329	3117.9516	2038.8447	1029.8864	881.1606	448.977
1	44.7274	2268.4432	3724.7433	3442.4533	2956.7563	1614.6579	905.4897	833.6311	615.7646
1	279.0043	1886.992	3161.3895	2282.4531	2694.9705	2315.1169	1430.5196	718.8912	717.9707
1	56.494	1208.2216	2533.9363	2375.5059	2547.9195	2141.7309	1411.468	1016.5037	601.5106
1	133.057	1562.8555	2172.7457	2026.8463	2294.3205	1827.0599	1421.1621	946.0587	527.5265

NS Lemon Sole: BTS & IBTS Q3; Last group is NOT a plus-group. Calculations made on 16/04/2025 at 11:33:00									
1	306.0939	1213.4921	2598.7277	2164.3356	1448.6414	1547.7858	1520.2883	1051.4339	688.566
1	620.6674	1587.8812	2402.3557	2324.5036	1548.783	1429.5389	1269.4907	963.1716	1040.8714
1	395.6743	1456.8728	3439.8418	2360.7126	2332.9702	1293.6343	907.65	898.6348	633.6272
1	710.3476	1660.5641	2154.8574	1728.4884	2186.2008	1204.5093	891.9572	815.8521	724.4858
1	477.4506	1746.5884	2491.5979	2479.9819	1985.8491	1514.7877	774.447	764.2213	390.1758
1	249.2408	1336.507	2500.0763	2990.1023	2386.5106	1648.5672	1038.538	431.4828	433.1615

Table 9.3.3. Lemon sole in Subarea 4 and divisions 3.a and 7.d. Biological sampling for IBTS Q1 survey (in numbers of individuals), indicating low sampling numbers in the most recent years.

Year/Age	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
2007	0	5	18	25	38	29	6	5	0	0	3	0	0	0	0	0	0	0
2008	0	15	49	35	35	13	36	13	8	6	3	22	1	1	3	0	1	0
2009	2	20	30	14	4	4	3	17	1	2	0	1	14	0	0	0	0	3
2010	3	42	51	47	9	2	2	0	0	0	0	0	0	0	0	0	0	0
2011	0	11	30	25	21	5	0	1	0	0	0	0	0	0	0	0	0	0
2012	0	40	60	26	15	2	1	0	0	0	0	0	0	0	0	0	0	0
2013	2	30	61	40	19	4	0	1	0	0	0	0	0	0	0	0	0	0
2014	1	29	64	32	4	2	0	0	0	0	0	0	0	0	0	0	0	0
2015	0	11	77	42	12	3	1	0	0	0	0	0	0	0	0	0	0	0
2016	0	52	121	49	10	1	1	0	0	0	0	0	0	0	0	0	0	0
2017	0	22	70	51	21	4	1	0	0	0	0	0	0	0	0	0	0	0
2018	1	32	72	38	10	5	1	0	0	0	0	0	0	0	0	0	0	0
2019	0	13	45	27	11	3	2	1	0	0	0	0	0	0	0	0	0	0
2020	0	18	55	24	6	0	0	0	0	0	0	0	0	0	0	0	0	0
2021	0	7	40	33	13	5	0	2	0	0	0	0	0	0	0	0	0	0
2022	0	8	21	10	7	1	0	0	0	0	0	0	0	0	0	0	0	0
2023	0	8	8	4	0	0	0	0	0	0	0	0	0	0	0	0	0	0
2024	0	0	7	13	9	4	4	5	2	0	0	0	0	0	0	0	0	0
2025	0	1	13	23	13	4	1	2	0	0	0	0	0	0	0	0	0	0

Table 9.3.4. Lemon sole in Subarea 4 and divisions 3.a and 7.d. Biological sampling in IBTS Q3 and BTS Q3 survey (in number of individuals).

Year/Age	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	22	27
2005	0	13	88	73	61	39	56	26	16	29	4	8	1	0	10	0	2	0	0	0	0	0	0
2006	27	19	86	129	91	97	49	84	24	14	45	10	9	1	0	13	0	1	0	2	0	1	0
2007	1	103	195	165	129	92	95	39	46	12	12	33	7	5	1	2	4	0	1	0	0	0	0
2008	5	37	305	234	134	117	68	90	43	32	12	3	32	0	4	1	1	4	0	0	0	0	0
2009	2	67	198	253	146	83	87	22	63	4	23	6	1	26	4	0	2	0	2	0	0	0	0
2010	1	42	59	86	75	90	67	78	37	48	7	25	1	1	11	1	1	2	0	1	1	0	0

Year/Age	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	22	27
2011	1	11	225	186	94	119	95	47	80	21	53	6	9	1	3	17	0	0	0	0	0	0	0
2012	3	40	217	162	120	93	83	60	47	70	20	30	8	22	3	0	6	1	1	0	0	0	0
2013	0	0	11	68	125	87	95	104	65	27	49	4	35	2	11	3	0	15	1	1	0	2	0
2014	5	27	69	112	132	134	100	56	49	31	19	26	3	12	6	11	2	2	2	0	0	0	0
2015	0	5	112	168	138	135	90	56	51	39	29	11	10	5	6	2	1	1	0	0	0	0	1
2016	1	25	106	148	101	111	119	71	39	42	20	24	7	9	2	8	7	2	0	0	0	0	0
2017	0	11	114	108	114	108	107	73	59	38	21	7	13	4	4	0	1	1	0	0	0	0	0
2018	10	31	261	184	129	154	125	111	73	51	25	15	8	9	0	5	3	1	0	1	0	0	0
2019	4	74	239	194	102	85	96	110	71	48	41	19	10	6	2	3	4	1	0	0	0	0	0
2020	12	225	225	155	99	70	79	72	60	65	40	24	22	9	4	4	5	3	1	0	1	0	0
2021	3	79	147	260	137	107	69	63	60	42	32	34	26	7	6	7	0	1	0	0	0	0	0
2022	3	107	226	132	107	121	67	57	56	50	43	24	24	12	9	1	0	1	1	0	0	0	0
2023	1	60	185	204	155	112	94	46	48	25	26	16	10	7	6	5	0	0	0	0	0	0	0
2024	0	32	122	149	156	135	106	70	29	25	20	10	8	5	6	3	0	1	0	0	0	0	0

Table 9.4.1. Lemon sole in Subarea 4 and divisions 3.a and 7.d. SURBAR stock summary. Mortality Z is given as the mean total mortality over ages 3-5, while SSB and recruitment at age 1 are mean-standardised relative indices. Each estimate is given with lower and upper bounds of a 90% confidence interval.

Year	z.low	z	z.high	ssb.low	ssb	ssb.high	rec.low	rec	rec.high
2005	-0.082	0.176	0.431	0.715	0.934	1.193	0.501	0.731	1.006
2006	-0.05	0.176	0.411	0.779	0.98	1.222	0.552	0.786	1.079
2007	0.162	0.396	0.64	0.802	1.019	1.275	0.804	1.168	1.652
2008	0.143	0.359	0.586	0.682	0.851	1.033	0.605	0.841	1.133
2009	-0.213	-0.01	0.192	0.584	0.715	0.871	0.607	0.832	1.094
2010	-0.186	0.017	0.227	0.766	0.947	1.159	0.819	1.122	1.467
2011	-0.07	0.129	0.341	0.931	1.157	1.422	0.864	1.197	1.611
2012	0.038	0.26	0.481	1.024	1.271	1.543	0.834	1.157	1.539
2013	-0.053	0.154	0.357	0.974	1.194	1.456	0.646	0.891	1.185
2014	-0.042	0.171	0.385	1.029	1.267	1.55	0.802	1.083	1.43
2015	-0.131	0.074	0.277	1.018	1.284	1.591	0.514	0.717	0.943
2016	-0.018	0.197	0.415	1.101	1.385	1.701	0.633	0.884	1.171

Year	z.low	z	z.high	ssb.low	ssb	ssb.high	rec.low	rec	rec.high
2017	0	0.208	0.43	1.003	1.279	1.584	0.569	0.793	1.057
2018	-0.097	0.109	0.314	0.883	1.128	1.398	0.755	1.043	1.38
2019	-0.122	0.098	0.308	0.877	1.096	1.341	0.716	1.006	1.352
2020	0.016	0.227	0.455	0.863	1.083	1.341	0.933	1.342	1.883
2021	0.012	0.223	0.434	0.752	0.94	1.158	1.082	1.589	2.215
2022	0.071	0.283	0.493	0.695	0.864	1.061	1.022	1.616	2.377
2023	0.117	0.388	0.638	0.612	0.769	0.946	0.443	0.779	1.274
2024	0.185	0.298	0.413	0.489	0.629	0.795	0.371	0.962	1.92

Table 9.5.1. Lemon sole in Subarea 4 and divisions 3.a and 7.d. Output from LBI analyses. Green shows indicators that are met or exceeded, while red shows indicators that are not met. Length data for 2013 are excluded from the analyses.

Year	Conservation		Optimising yield		MSY	
	Lc/Lmat	L25/Lmat	Lmax5%/Linf	Pmega	Lmean/Lopt	Lmean/L(F=M)
	>1	>1	>0.8	>30%	~1 (>0.9)	>=1
2002	1.838	1.728	1.001	0.588	1.225	1.089
2003	1.103	1.654	0.997	0.481	1.074	1.302
2004	1.691	1.801	1.001	0.609	1.202	1.128
2005	1.838	1.801	0.910	0.383	1.126	1.001
2006	1.838	1.801	0.962	0.555	1.178	1.047
2007	1.838	1.801	0.975	0.501	1.151	1.023
2008	1.397	1.654	0.996	0.477	1.105	1.170
2009	0.809	1.654	0.994	0.479	1.079	1.531
2010	1.691	1.728	1.005	0.518	1.179	1.107
2011	0.956	1.287	0.959	0.285	0.938	1.227
2012	1.397	1.434	0.948	0.267	1.019	1.078
2013	NA	NA	NA	NA	NA	NA
2014	1.103	1.434	0.988	0.325	0.986	1.195
2015	1.397	1.507	0.995	0.284	1.036	1.096
2016	1.397	1.507	1.005	0.449	1.092	1.156
2017	1.397	1.507	1.023	0.499	1.135	1.201
2018	1.985	1.875	1.076	0.698	1.291	1.090

	Conservation		Optimising yield		MSY	
2019	1.397	1.434	1.024	0.433	1.119	1.185
2020	1.103	1.581	1.034	0.518	1.081	1.310
2021	1.397	1.507	0.997	0.402	1.079	1.142
2022	1.397	1.507	0.995	0.374	1.080	1.143
2023	1.397	1.507	0.963	0.228	0.992	1.050
2024	1.397	1.581	1.028	0.445	1.094	1.158

Table 9.5.2. Lemon sole in Subarea 4 and divisions 3.a and 7.d. Catch, stock size indicator and harvest rate used for the application of the *chr* rule. Note the year 2013 is not used for the calculation of the harvest rate. Harvest rate values from 2007-2022 used for the calculation of the $HR_{MSY\text{ proxy}}$ (excluding 2013).

Year	Catch (C)	Index (I)	Harvest rate (C/I)
2007	4919	1.019	4827
2008	5051	0.851	5935
2009	4401	0.715	6156
2010	3907	0.947	4126
2011	5055	1.157	4369
2012	6560	1.271	5162
2014	5335	1.267	4211
2015	5116	1.284	3985
2016	5000	1.385	3610
2017	3966	1.279	3101
2018	3376	1.128	2993
2019	3878	1.096	3538
2020	3044	1.083	2811
2021	2774	0.94	2951
2022	1961	0.864	2270
2023	1891	0.769	2459
2024	1802	0.629	2865

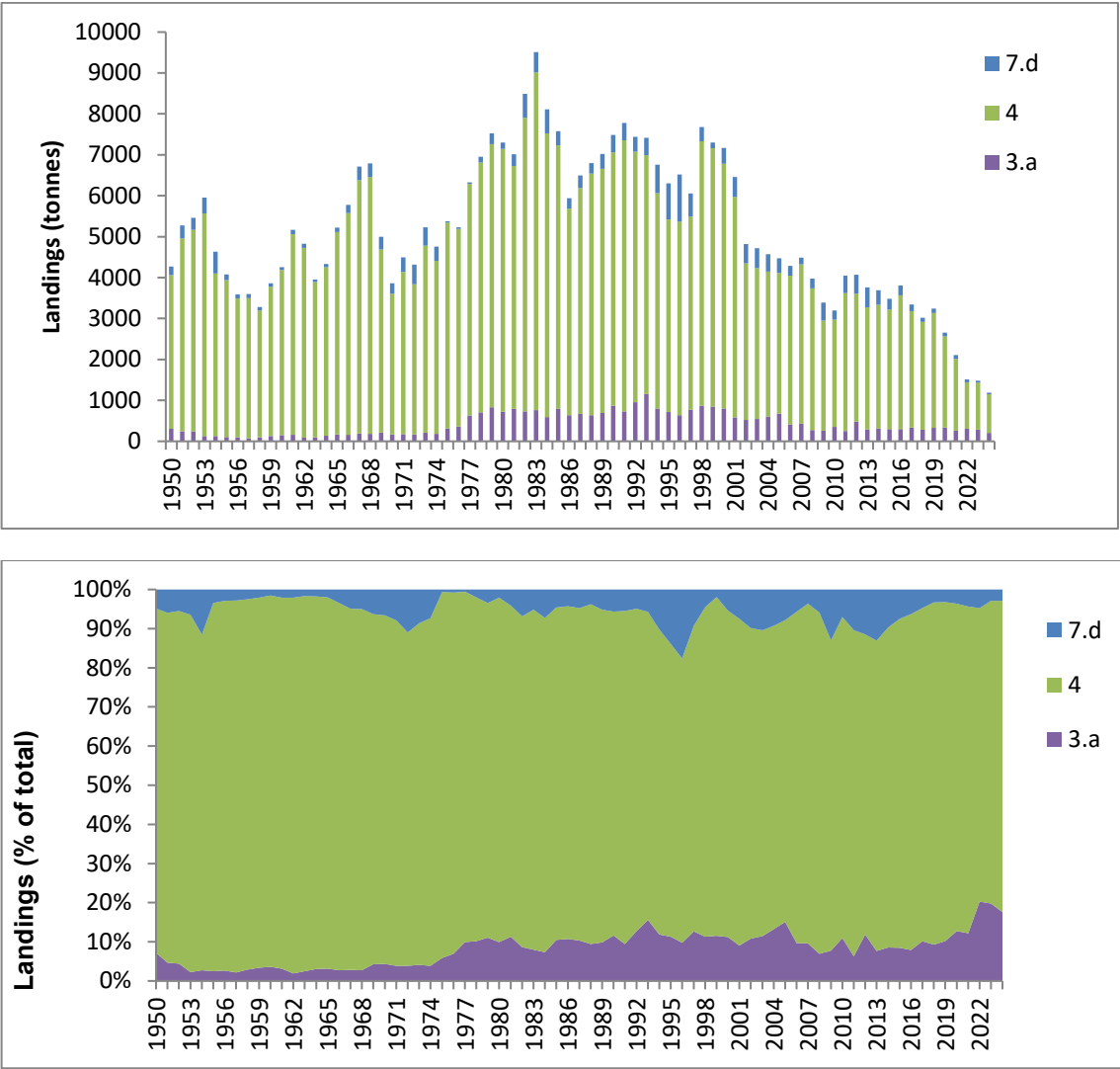


Figure 9.2.1. Lemon sole in Subarea 4 and divisions 3.a and 7.d. Officially-reported landings of lemon sole by area in the greater North Sea. Upper plot: landings by area in tonnes. Lower plot: landings by area as a percentage of the total landings.

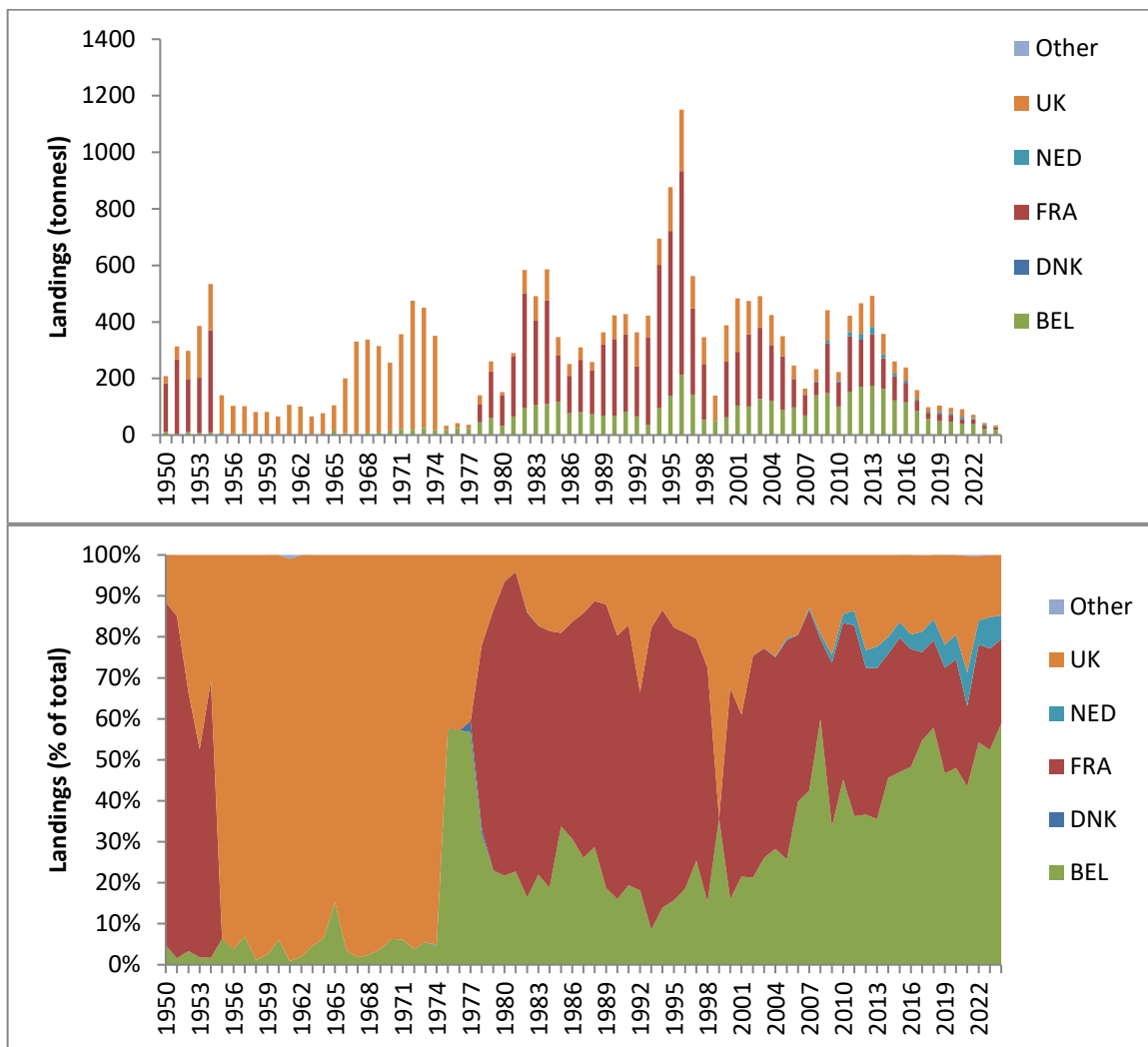


Figure 9.2.2. Lemon sole in Subarea 4 and divisions 3.a and 7.d. Official landings of lemon sole in Division 7.d by country. Upper plot: landings by country in tonnes. Lower plot: landings by country as a percentage of the total landings in Division 7.d.

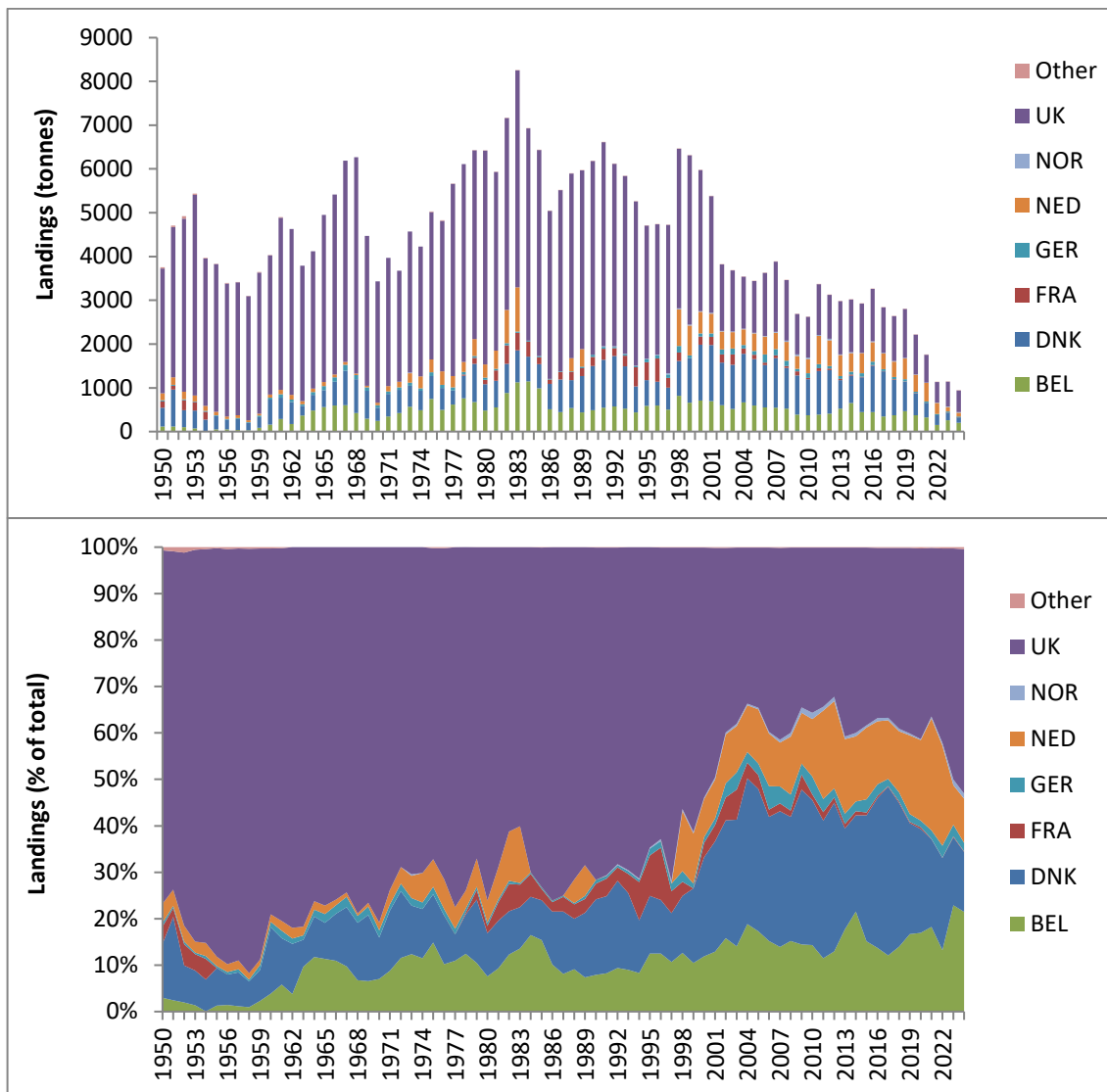


Figure 9.2.3. Lemon sole in Subarea 4 and divisions 3.a and 7.d. Official landings of lemon sole in Subarea 4 by country. Upper plot: landings by country in tonnes. Lower plot: landings by country as a percentage of the total landings in Subarea 4.

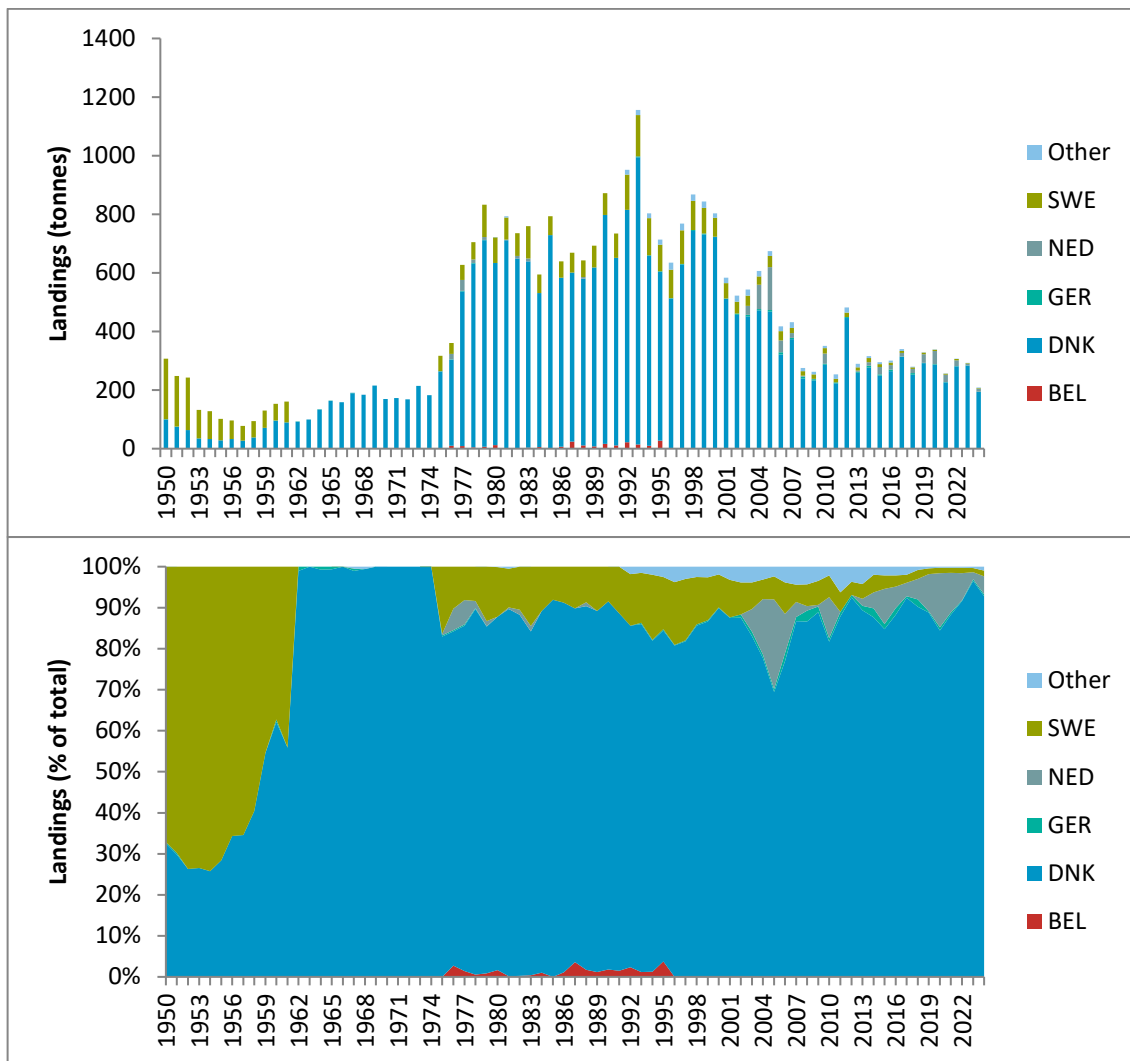


Figure 9.2.4. Lemon sole in Subarea 4 and divisions 3.a and 7.d. Official landings of lemon sole in Division 3.a by country. Upper plot: landings by country in tonnes. Lower plot: landings by country as a percentage of the total landings in Division 3.a.

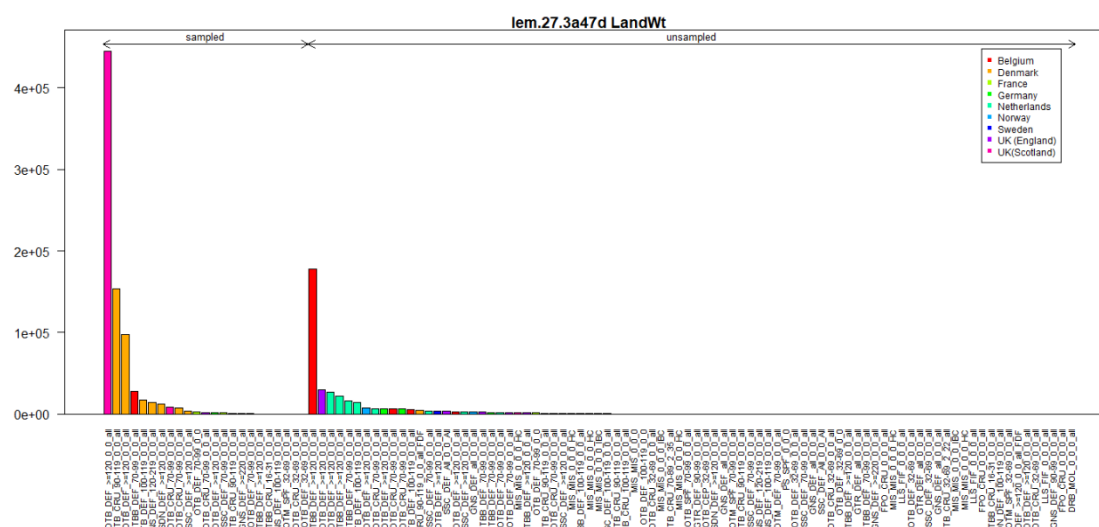


Figure 9.2.5. Lemon sole in Subarea 4 and divisions 3.a and 7.d. InterCatch summary plots. Sampled and unsampled fleets for landings yield estimation (tonnes).

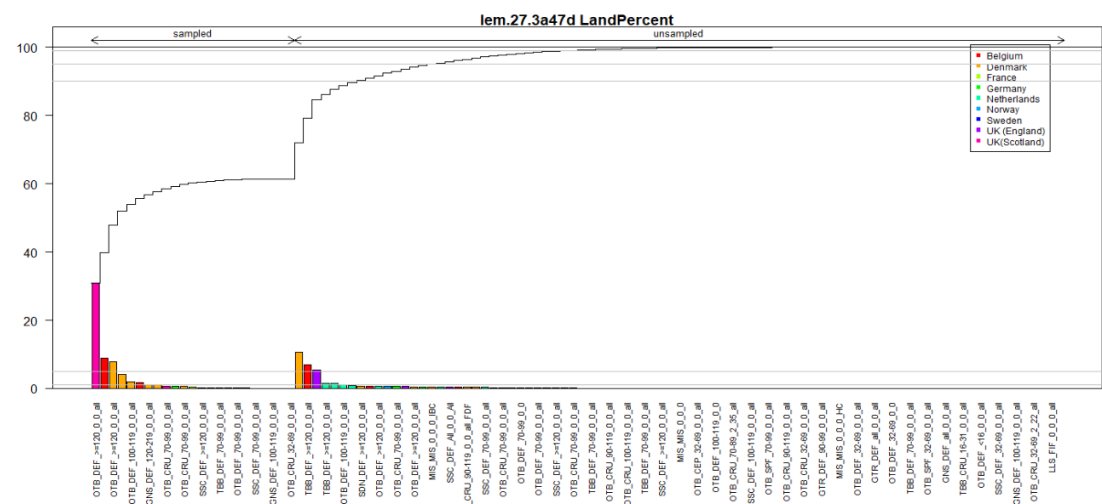


Figure 9.2.6. Lemon sole in Subarea 4 and divisions 3.a and 7.d. InterCatch summary plots. Sampled and unsampled fleets for landings yield estimation (cumulative contribution).

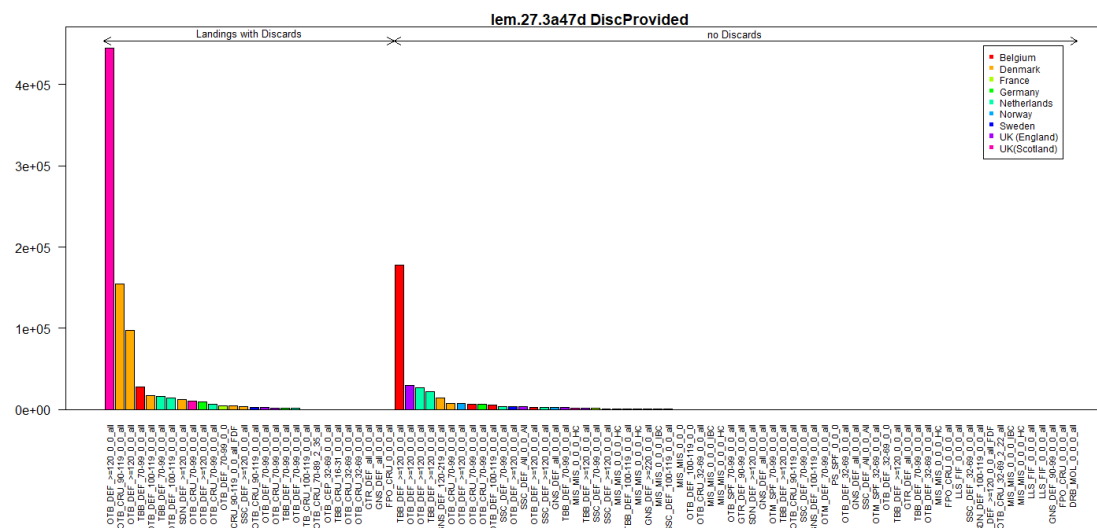


Figure 9.2.7. Lemon sole in Subarea 4 and divisions 3.a and 7.d. InterCatch summary plots. Fleets provided with and without discard estimates.

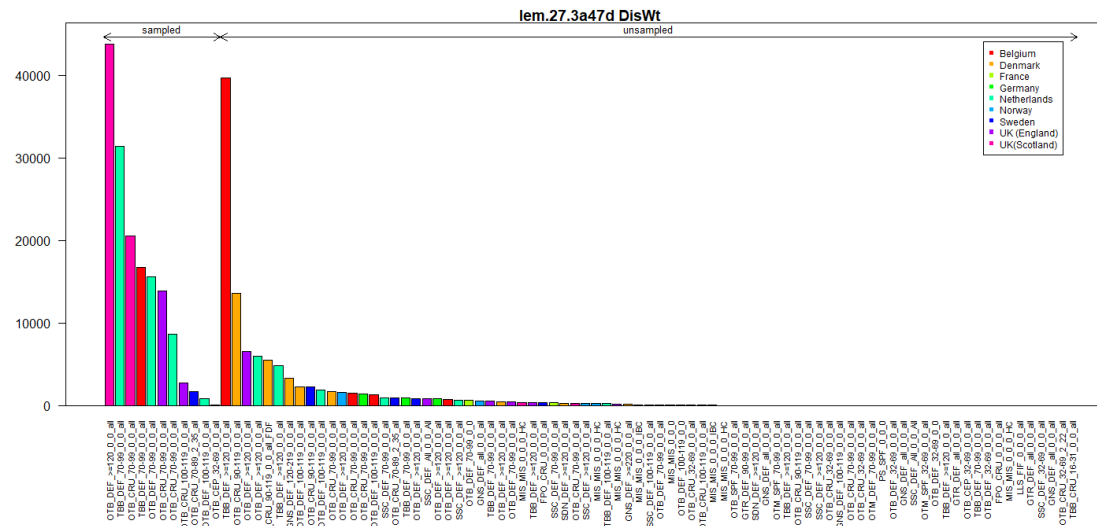


Figure 9.2.8. Lemon sole in Subarea 4 and divisions 3.a and 7.d. InterCatch summary plots. Sampled and unsampled fleets for discard yield estimation.

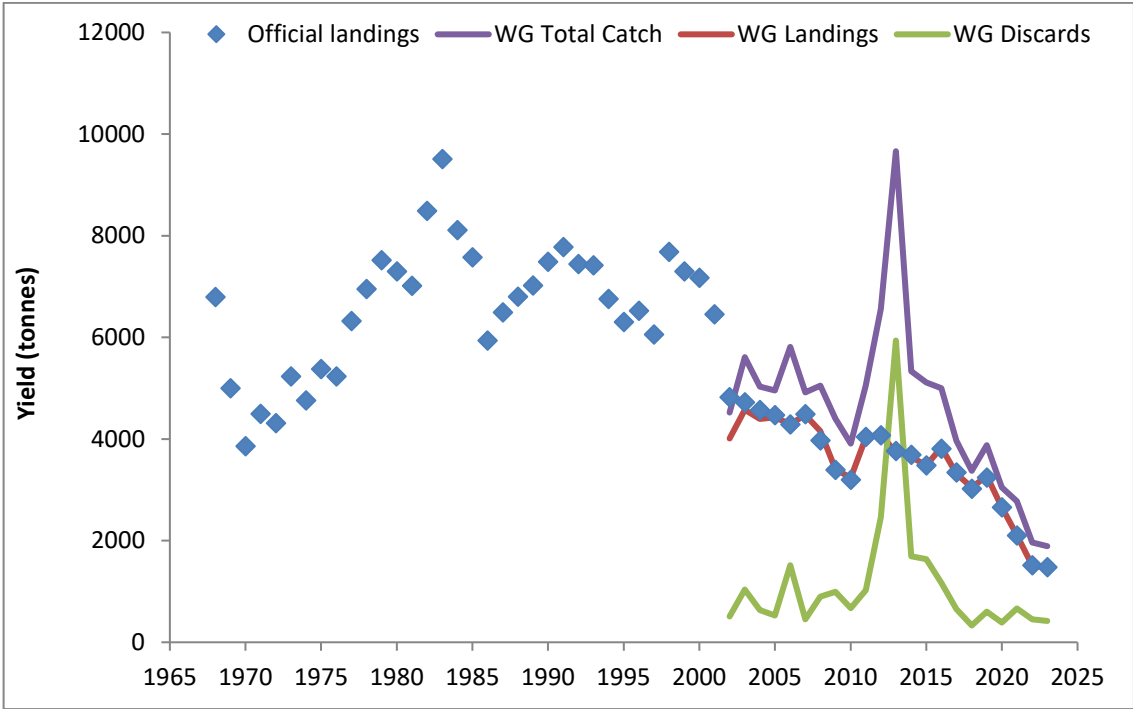


Figure 9.2.9. Lemon sole in Subarea 4 and divisions 3.a and 7.d. Time-series of official landings (dots) along with ICES estimates of total catch (purple line), landings (red line) and discards (green line).

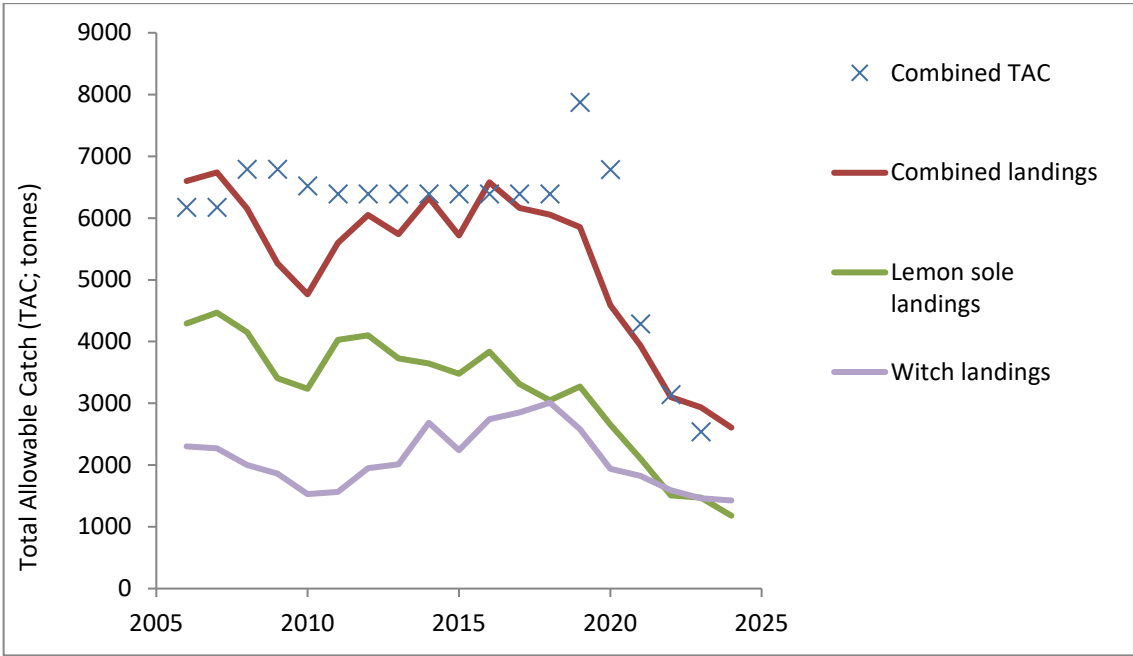


Figure 9.2.10. Lemon sole in Subarea 4 and divisions 3.a and 7.d. Time-series of ICES estimates of landings for lemon sole (green), witch (purple) and combined (red), along with the joint lemon sole-witch TAC (crosses). Since 2019, with implementation of TAC on catches rather than landings, respective WG estimates of catches are shown.

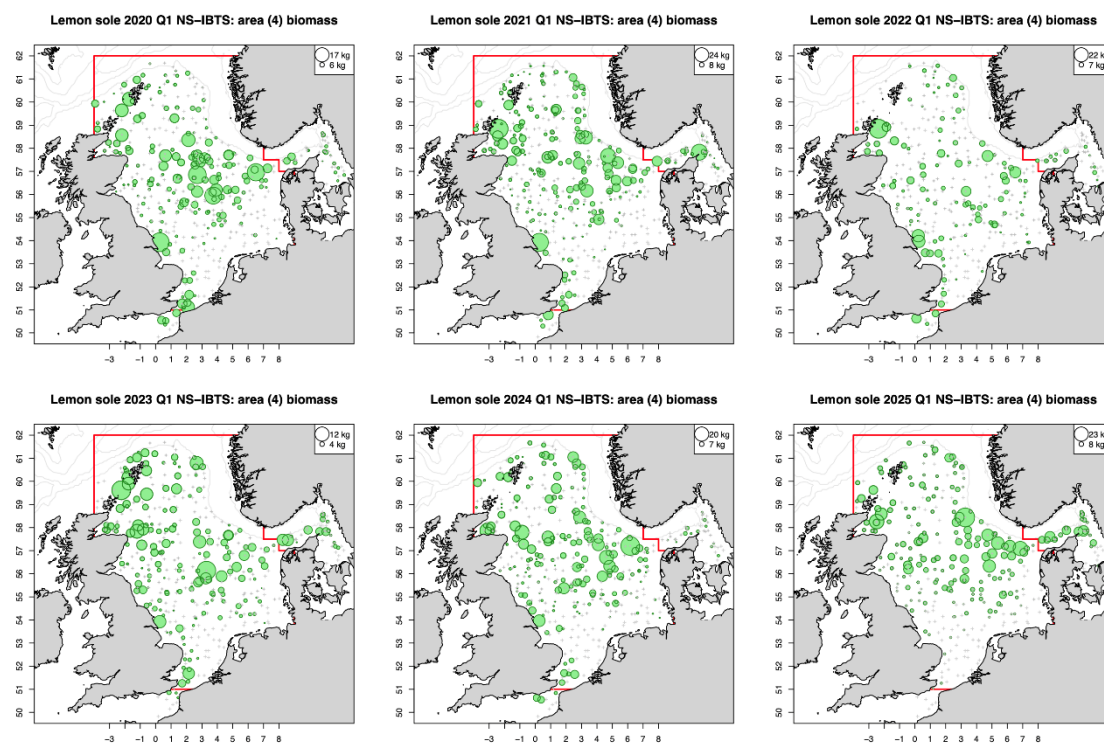


Figure 9.3.1a. Lemon sole in Subarea 4 and divisions 3.a and 7.d. Distribution of lemon sole in the North Sea derived from IBTS Q1 during 2020-2025. The sizes of the circles are proportional to the square root of the estimated weight of lemon sole caught in each haul.

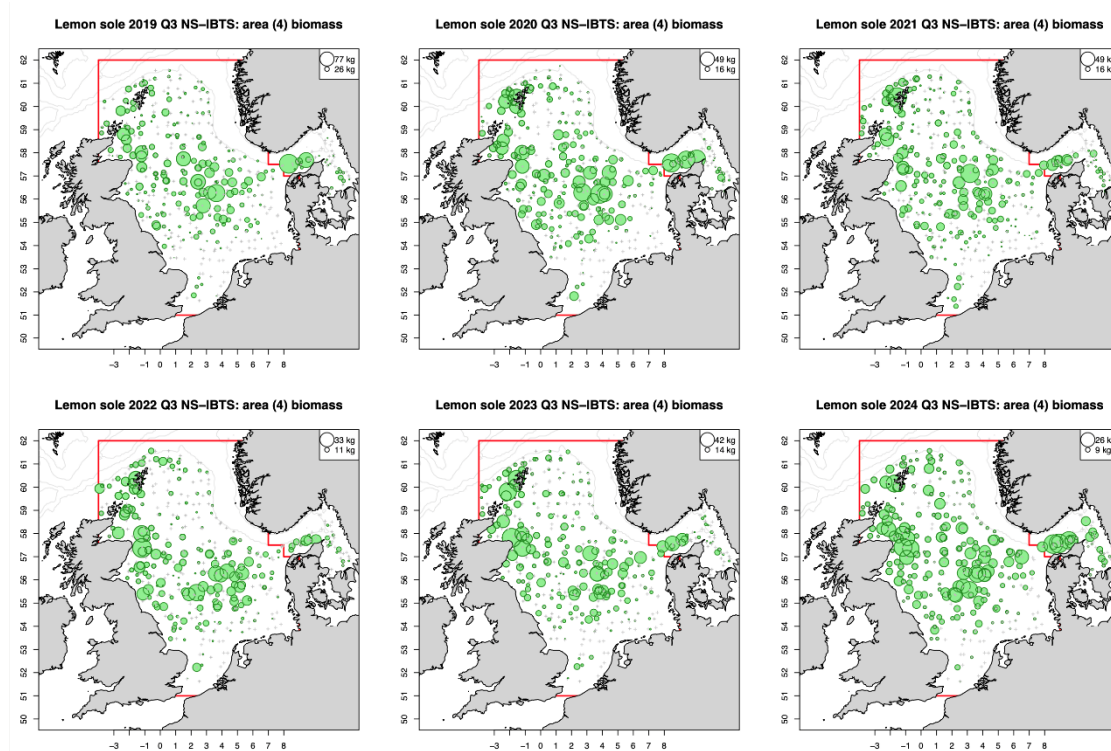


Figure 9.3.1b. Lemon sole in Subarea 4 and divisions 3.a and 7.d. Distribution of lemon sole in the North Sea derived from IBTS Q3 during 2019-2024. The sizes of the circles are proportional to the square root of the estimated weight of lemon sole caught in each haul.

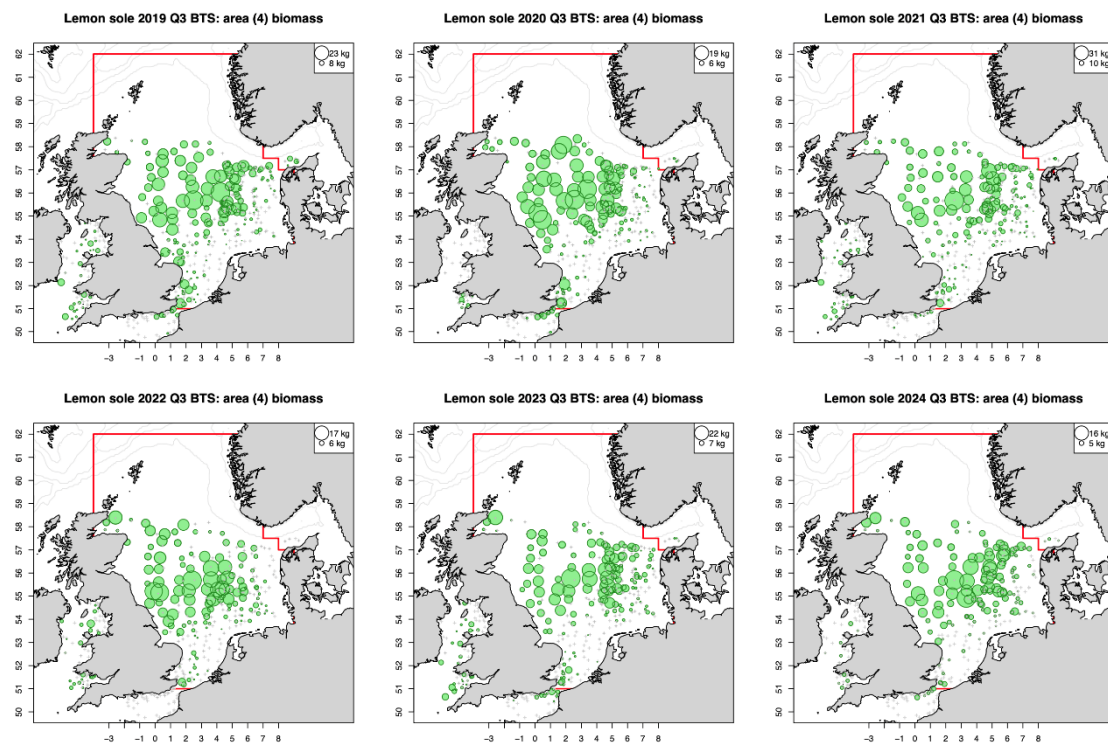


Figure 9.3.1c. Lemon sole in Subarea 4 and divisions 3.a and 7.d. Distribution of lemon sole in the North Sea derived from BTS Q3 during 2019-2024. The sizes of the circles are proportional to the square root of the estimated weight of lemon sole caught in each haul.

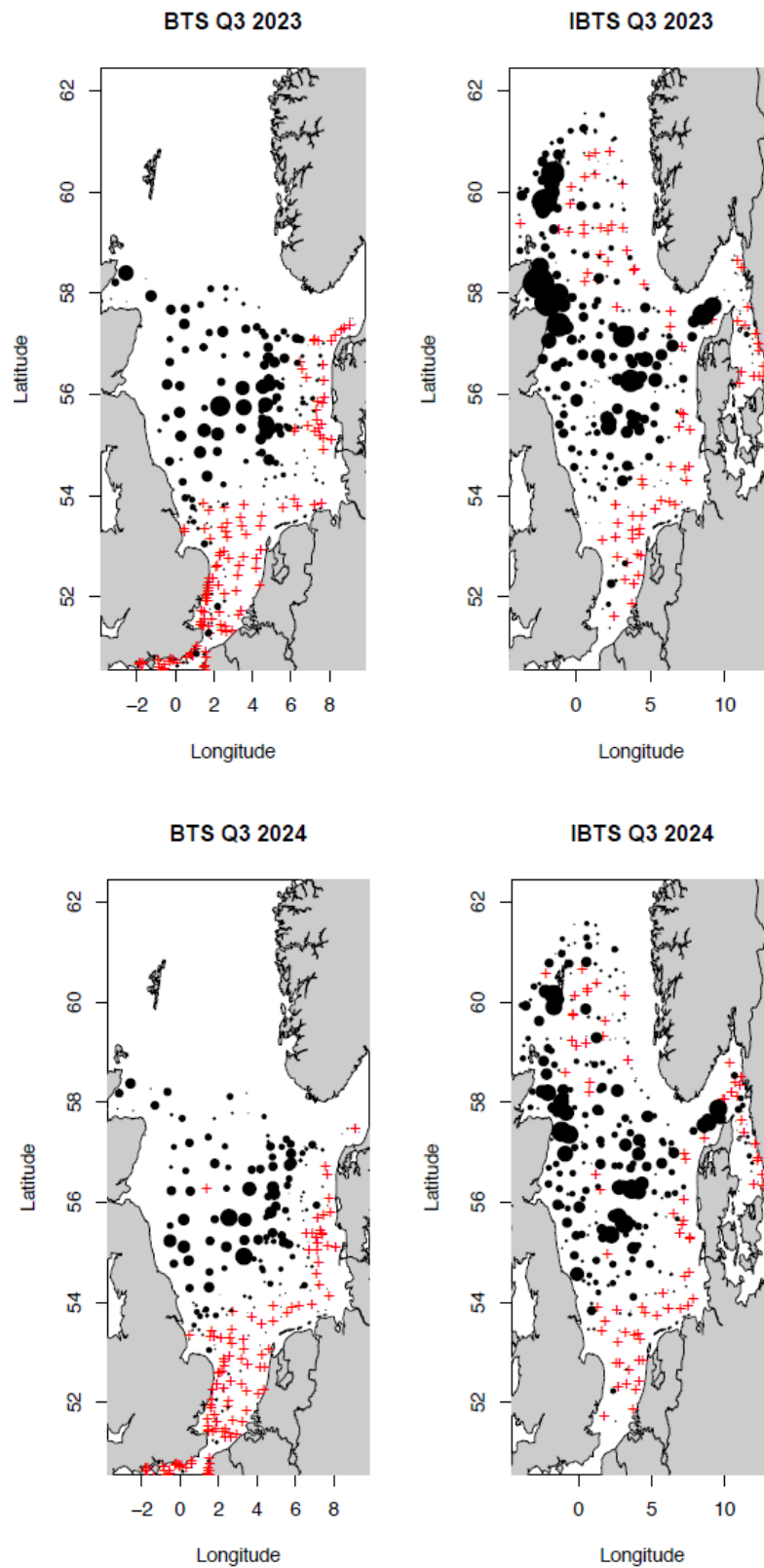


Figure 9.3.1d. Lemon sole in Subarea 4 and divisions 3.a and 7.d. Distribution of lemon sole in the North Sea derived from IBTS Q3 and BTS Q3 during 2023-2024. Red crosses indicating hauls with zero catches of lemon sole.

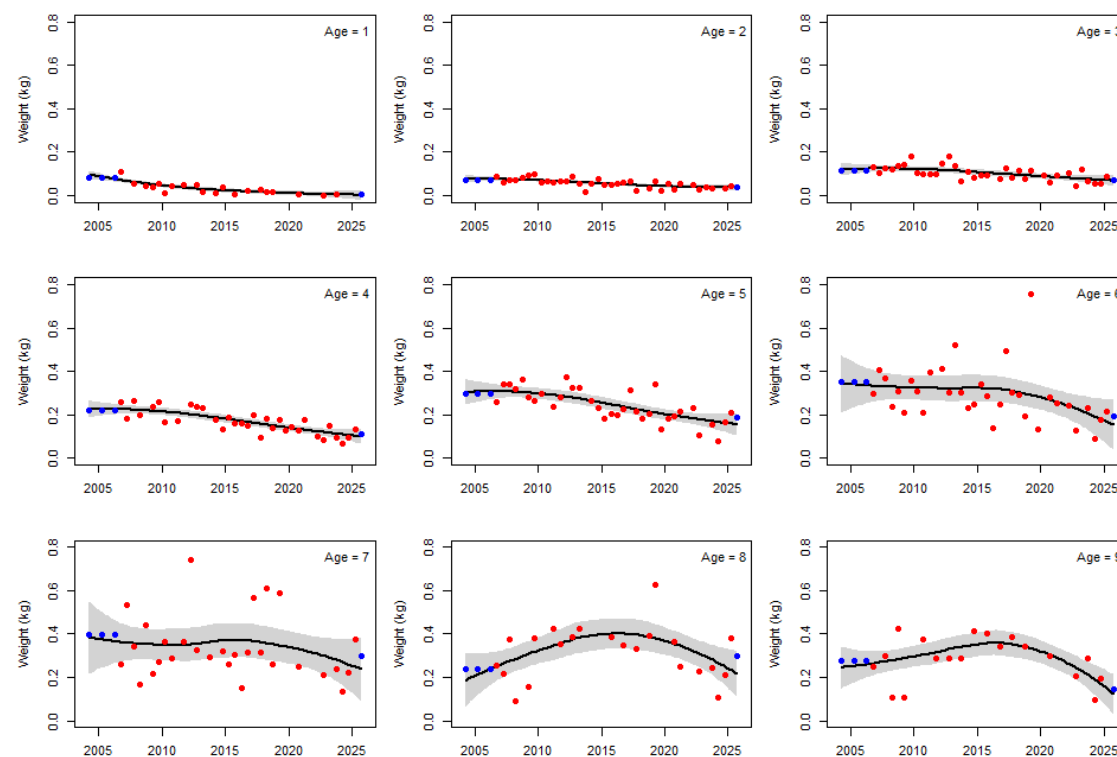


Figure 9.3.2. Lemon sole in Subarea 4 and divisions 3.a and 7.d. Time-series of mean weight-at-age estimates (red dots) from IBTS Q1 and Q3 surveys, summarised by a *loess* smoother (span = 1) for each year (the grey band gives a 95% confidence interval about the loess smoother). The blue dots show averages (of either the first or last two estimates), included to allow extrapolation to the start and end point of the survey indices.

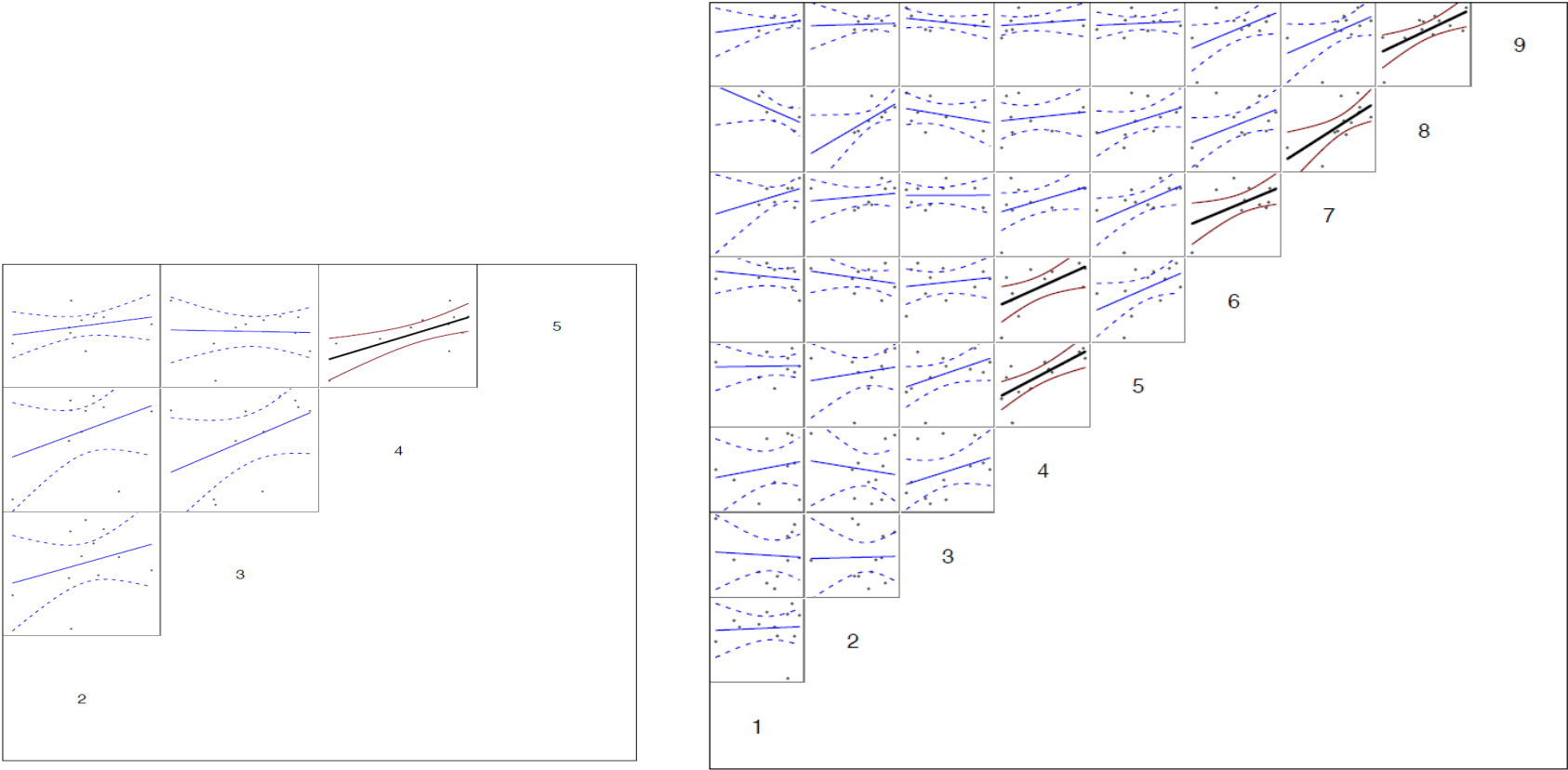


Figure 9.3.3. Lemon sole in Subarea 4 and divisions 3.a and 7.d. Bivariate scatterplots showing consistency in cohort-strength estimation, for Q1 (left: IBTS) and Q3 (right: IBTS and BTS).

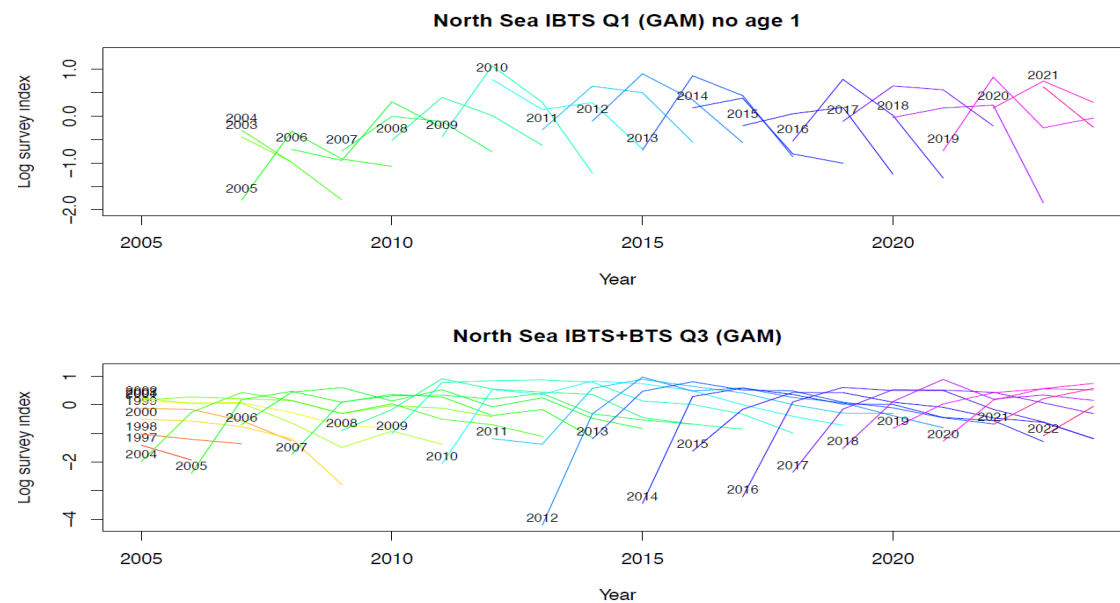


Figure 9.3.4. Lemon sole in Subarea 4 and divisions 3.a and 7.d. Survey catch curves for Q1 (upper: IBTS) and Q3 (lower: combined IBTS and BTS).

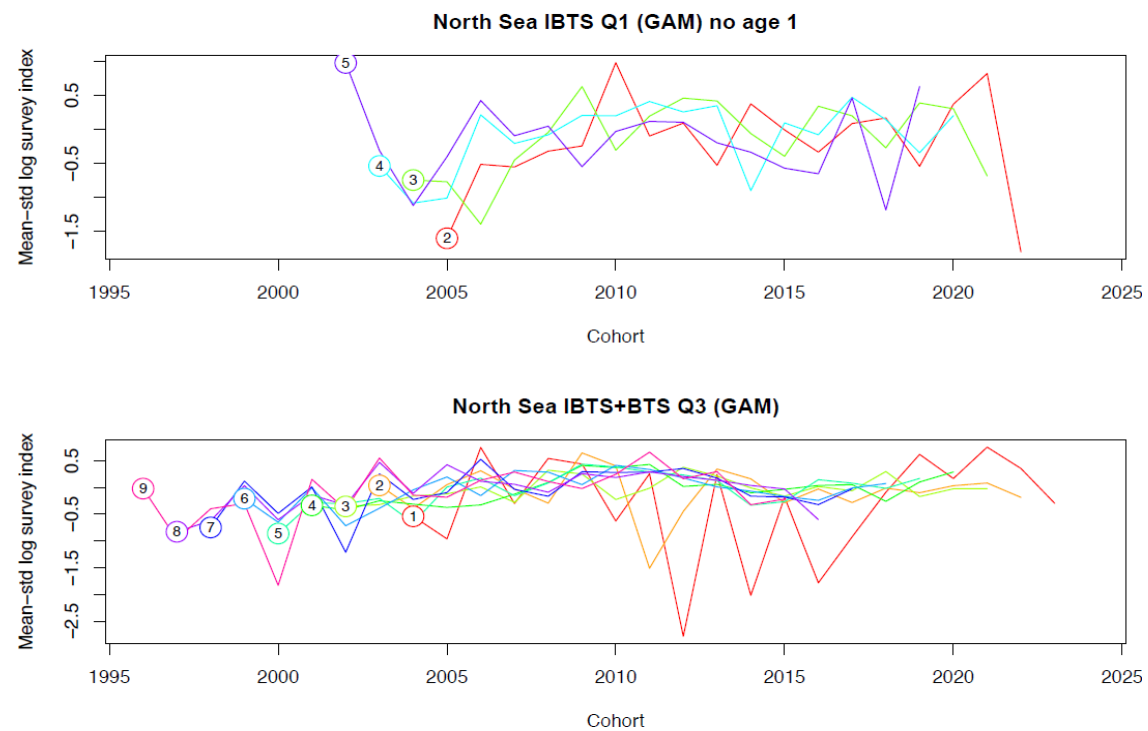


Figure 9.3.5. Lemon sole in Subarea 4 and divisions 3.a and 7.d. Survey indices by age, cohort and year for Q1 (top: IBTS) and Q3 (bottom: IBTS and BTS).

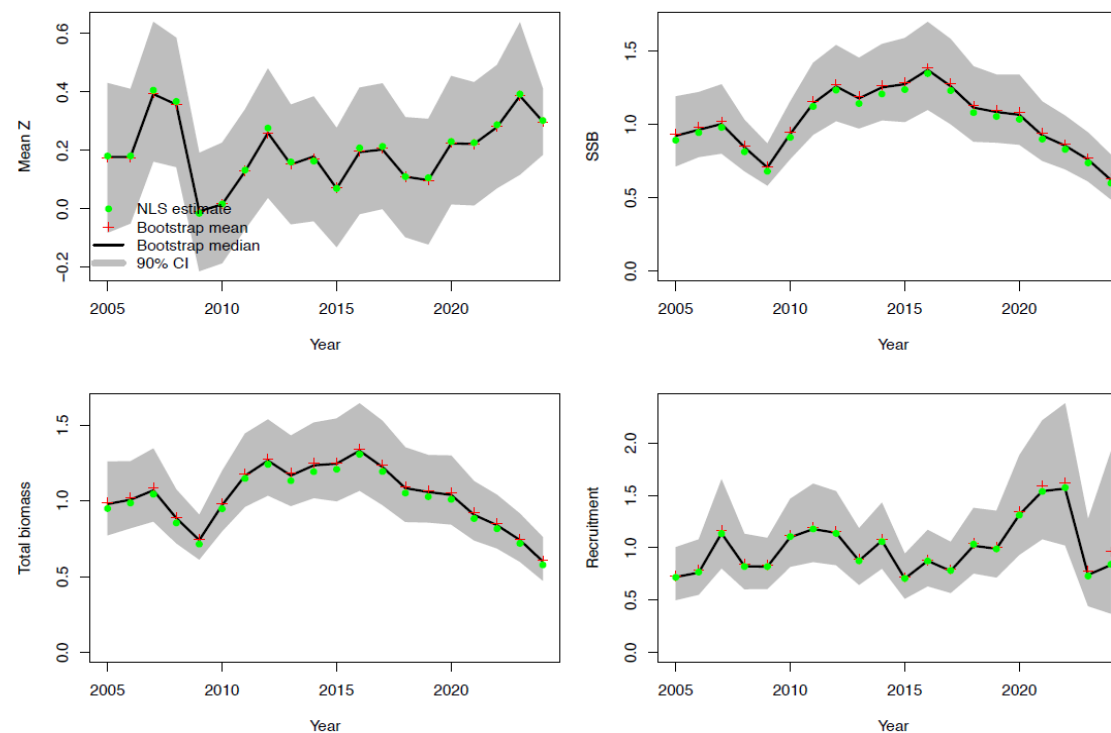


Figure 9.4.1. Lemon sole in Subarea 4 and divisions 3.a and 7.d. SURBAR stock summary (clockwise from top left: mean $Z(3-5)$, relative SSB, relative recruitment at age 1, relative total biomass). In each plot, the green dots give the nonlinear least-squares estimates, the red crosses give the uncertainty-estimation bootstrap mean, the black line gives the bootstrap median, and the grey band gives a 90% confidence interval about the media.

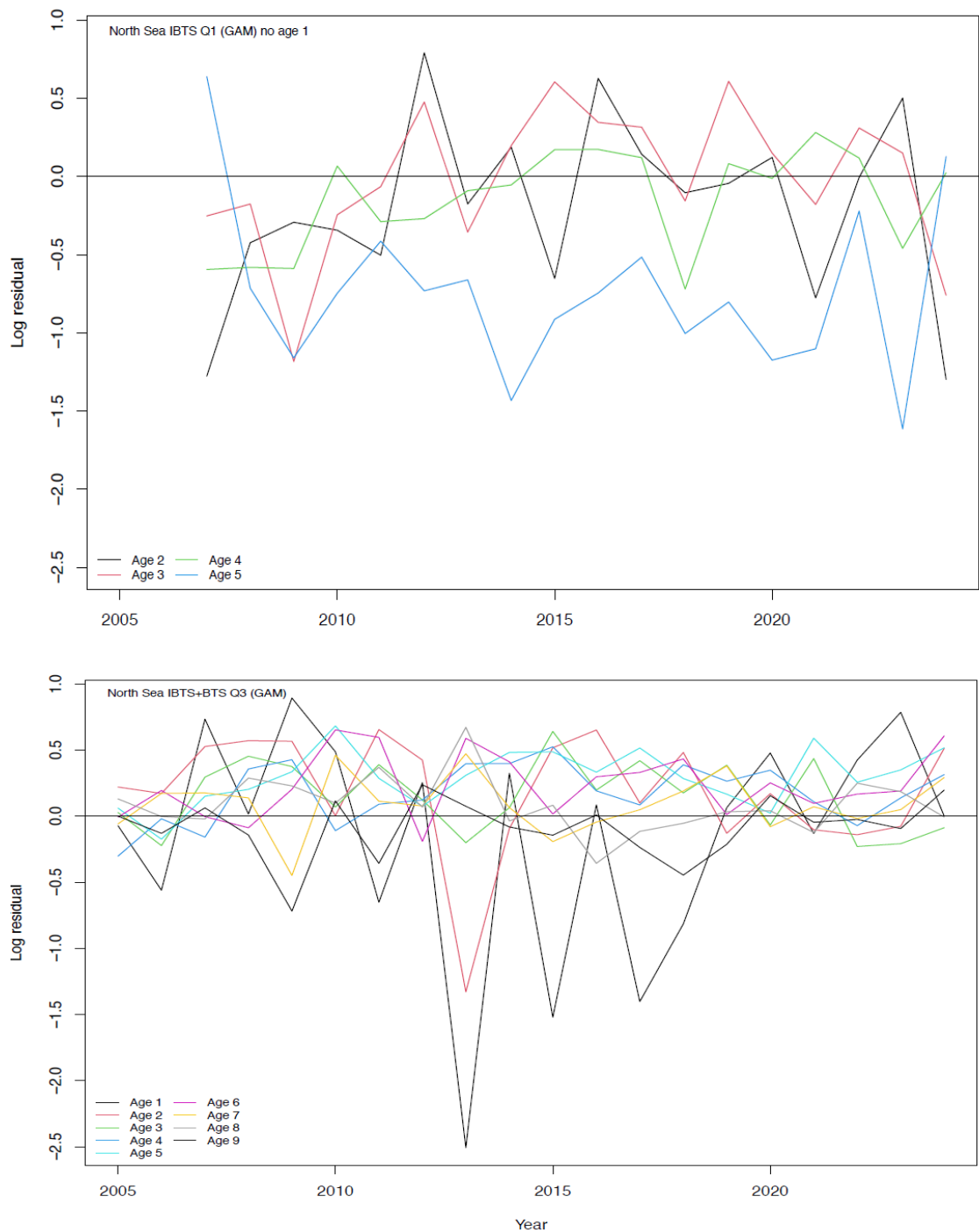


Figure 9.4.2. Lemon sole in Subarea 4 and divisions 3.a and 7.d. SURBAR residuals. Top: Log SURBAR residuals for Q1 (IBTS). Bottom: Log SURBAR residuals for Q3 (IBTS+BTS).

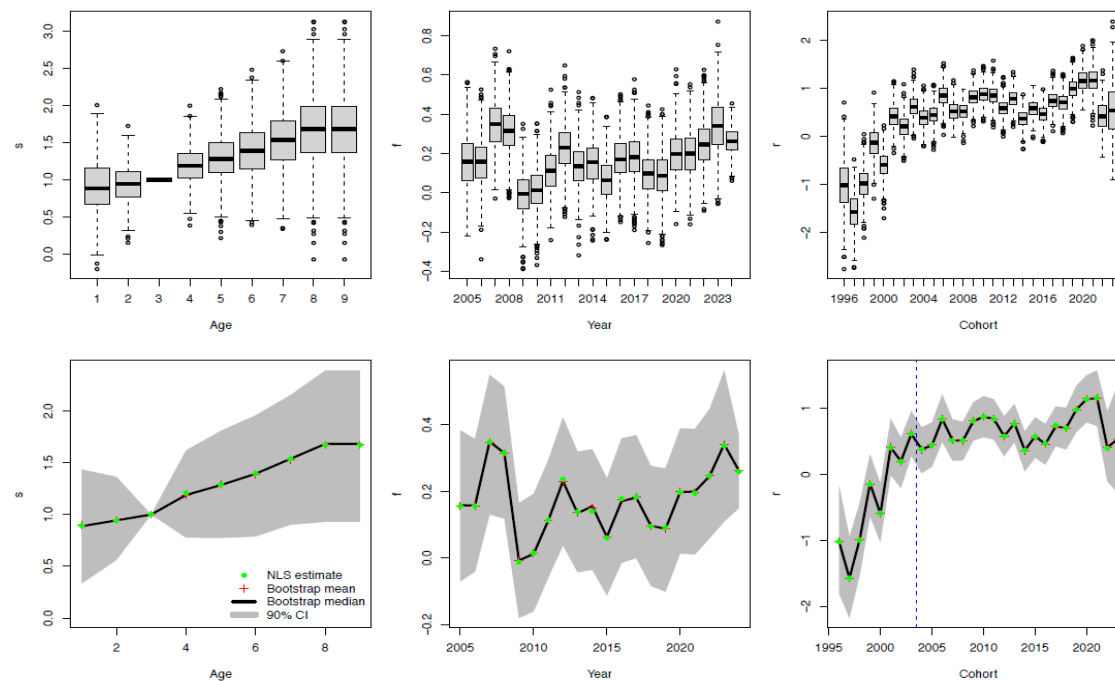


Figure 9.4.3. Lemon sole in Subarea 4 and divisions 3.a and 7.d. Estimated SURBAR parameters: age effects (s) and year effects (f) of total mortality, and cohort effects (r). Top: box-and-whisker plots of bootstrap distributions. Bottom: the green dots give the nonlinear least-squares estimates, the red crosses give the uncertainty-estimation bootstrap means, the black line gives the bootstrap median, and the grey band gives a 90% confidence interval about the median.

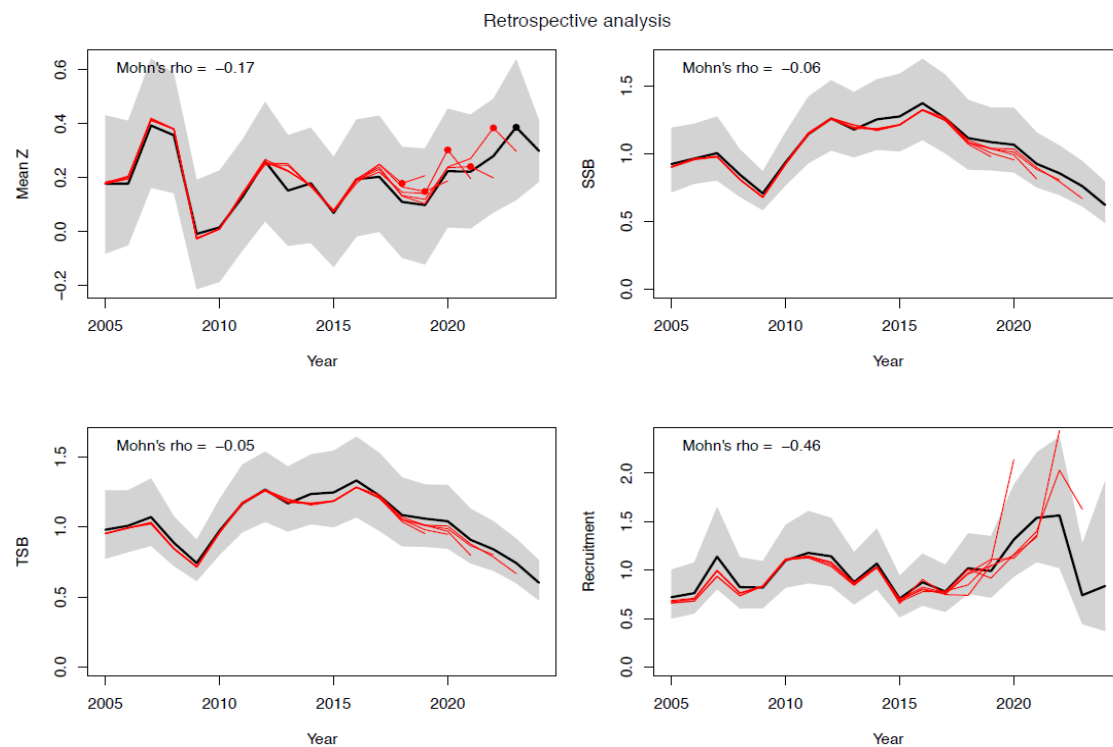


Figure 9.4.4. Lemon sole in Subarea 4 and divisions 3.a and 7.d. Retrospective SURBAR analysis (clockwise from top left: mean Z(3-5), relative SSB, relative total biomass, relative recruitment at age 1) with Mohn's rho included in each plot. Black lines give final-year estimates (with 90% confidence interval in grey), while red lines give the results of 5 retrospective peels.

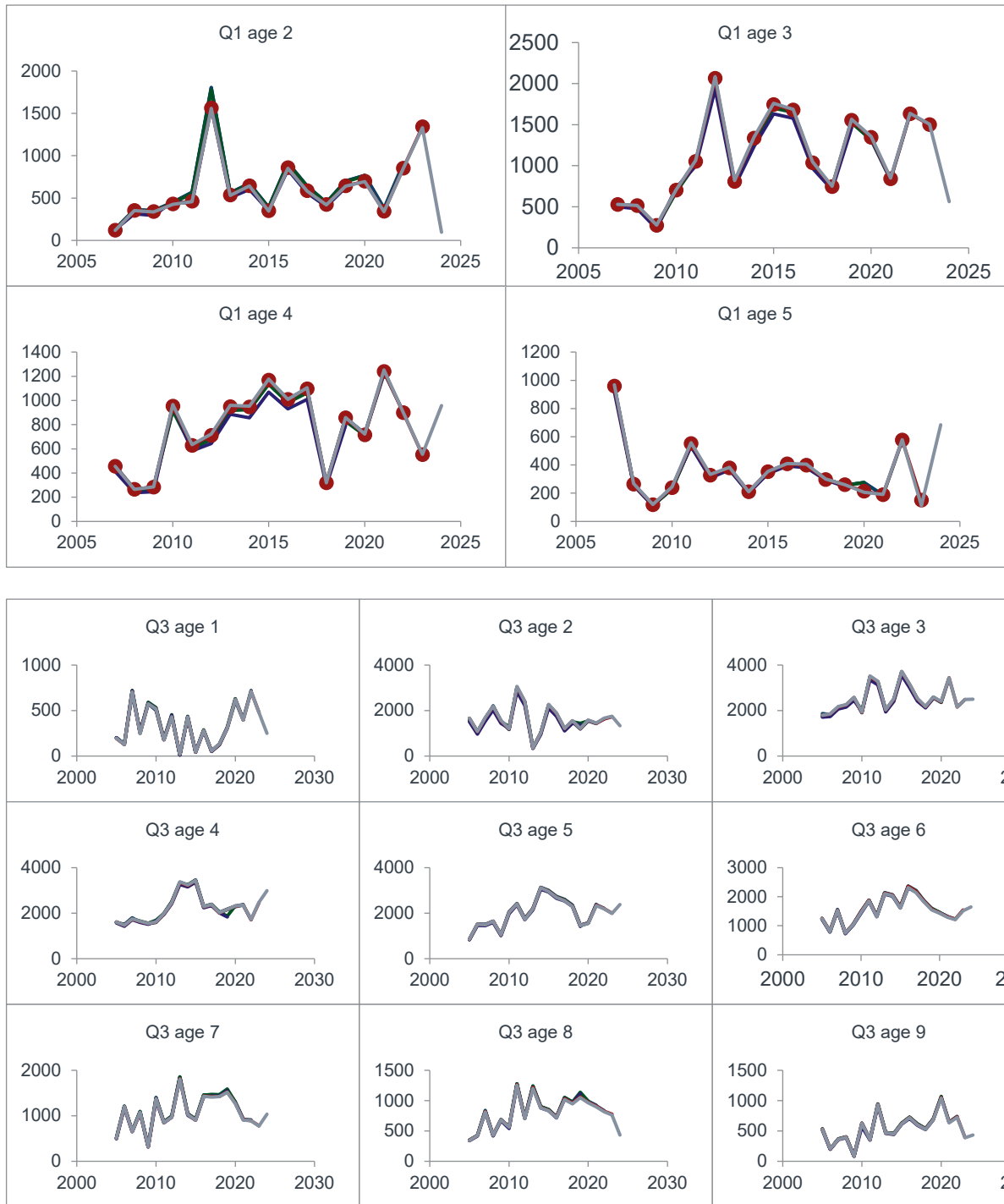


Figure 9.4.5. Lemon sole in Subarea 4 and divisions 3.a and 7.d. Historical retrospective comparisons of relative abundance indices from IBTS Q1 (top) and IBTS+BTS Q3 (bottom). The lines show survey indices from five successive meetings of WGNSSK.

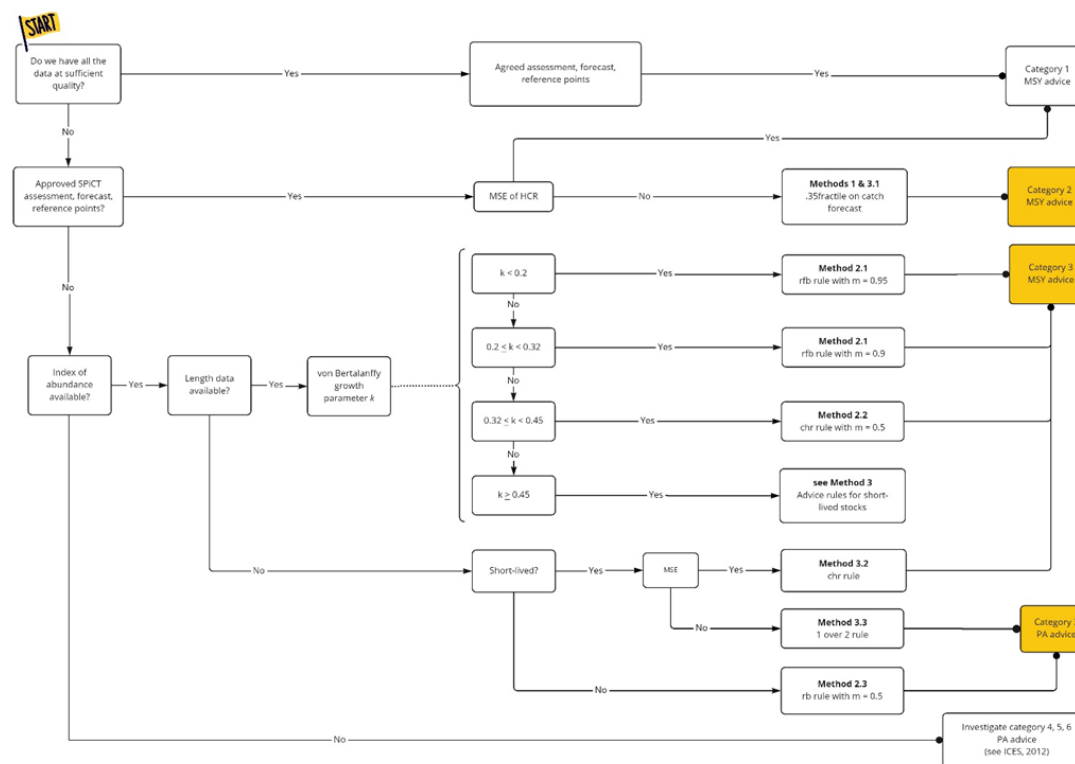


Figure 9.5.1. Lemon sole in Subarea 4 and divisions 3.a and 7.d. The WKLIFE DLS advice decision tree (ICES, 2022).

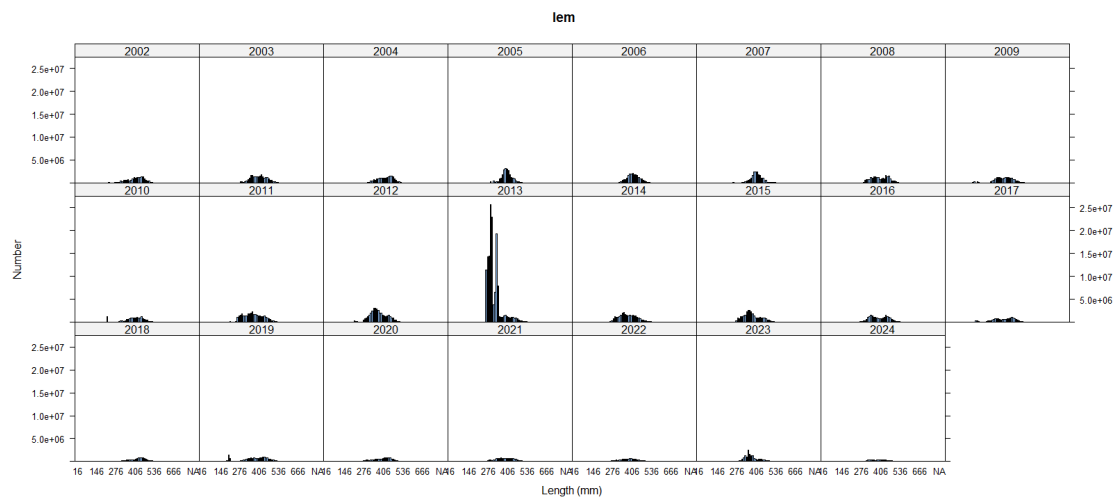


Figure 9.5.2. Lemon sole in Subarea 4 and divisions 3.a and 7.d. Length distributions in commercial catches (landing and discards) submitted to InterCatch, by year.

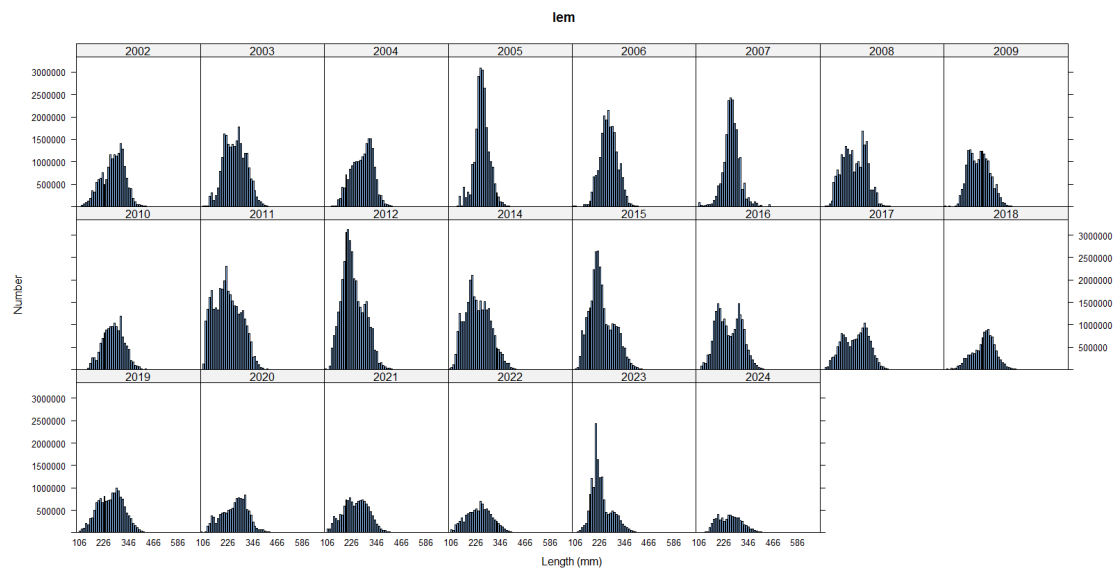


Figure 9.5.3. Lemon sole in Subarea 4 and divisions 3.a and 7.d. Length distributions in commercial catches (landing and discards) submitted to InterCatch, by year, with 2013 data removed due, and all fish <100 mm removed for all years for L_c calculation.

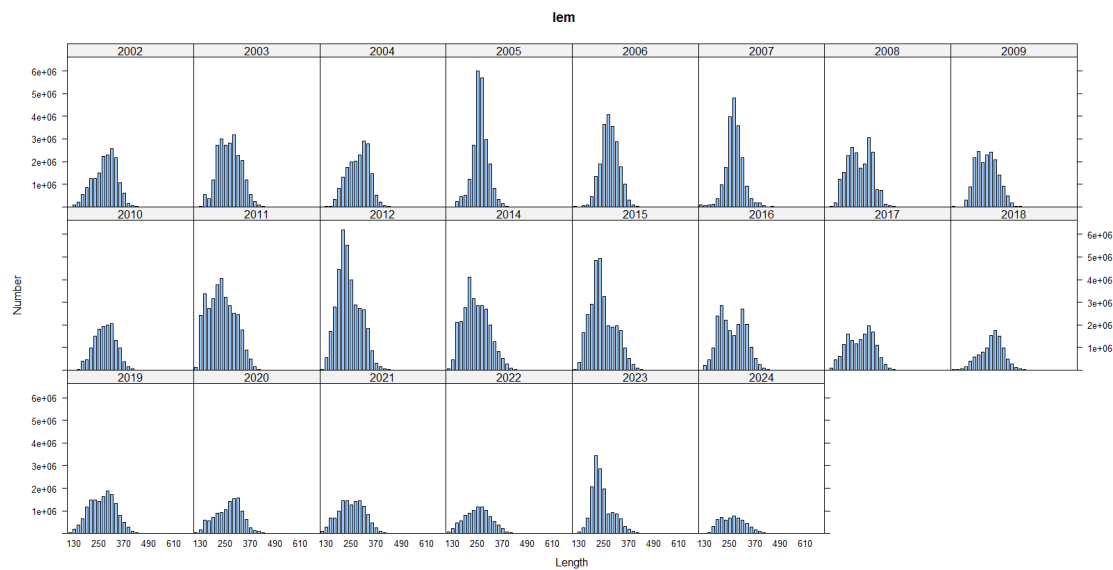


Figure 9.5.4. Lemon sole in Subarea 4 and divisions 3.a and 7.d. As for Figure 9.5.3, with bin widths doubled (to 20 mm).

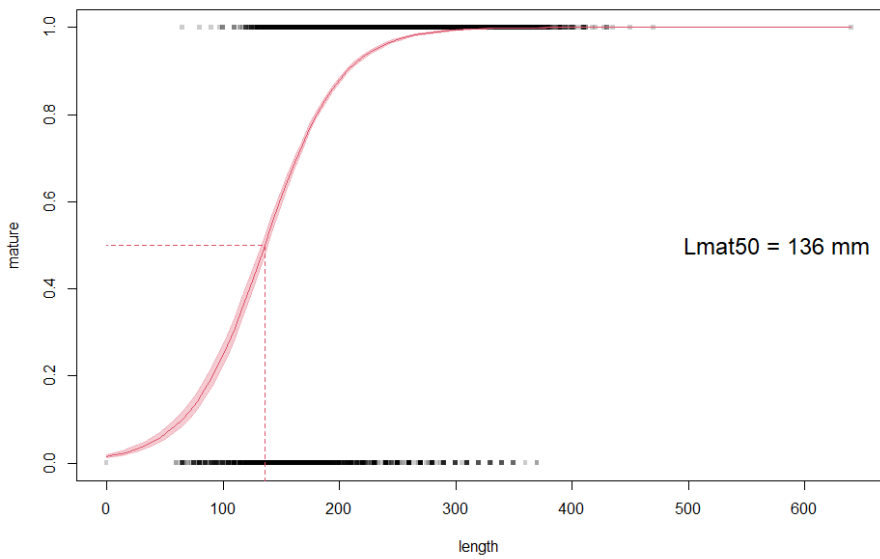


Figure 9.5.5. Lemon sole in Subarea 4 and divisions 3.a and 7.d. Fitted maturity-at-age estimates from Q1 (IBTS) and Q3 (IBTS & BTS) survey series, using maturity-length observations from all available years (2007-2024). Maturity indices (0 = not mature, 1 = mature) are shown as shaded dots. The solid red line gives the fitted maturity ogive with 95% confidence interval (red band), while dotted red lines highlight the length of 50% maturity ($L_{mat} = 136$ mm)

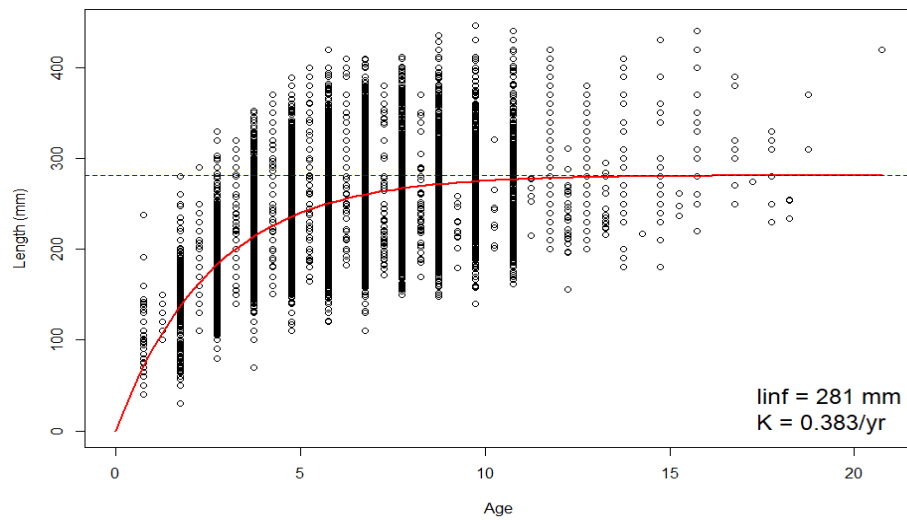


Figure 9.5.6. Lemon sole in Subarea 4 and divisions 3.a and 7.d. Length-at-age data from Q1 (IBTS) and Q3 (IBTS & BTS) survey series, using data from all available years (2005–2024). To account for seasons, Q1 lengths are plotted at $a + 0.25$, Q3 lengths at $a + 0.75$. The red line gives a fitted von Bertalanffy growth curve ($L_{\infty} = 281\text{m}$, $K = 0.383$, $t_0 = 0$).

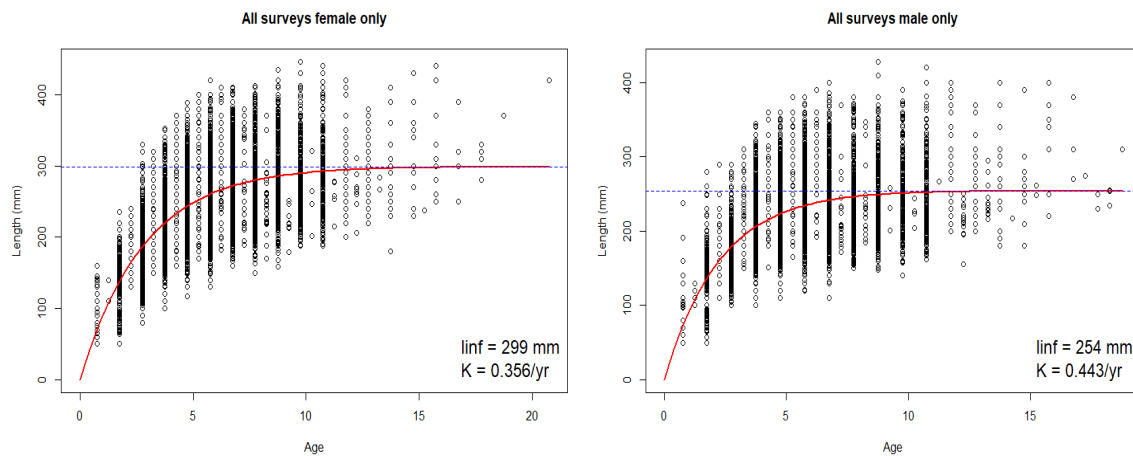


Figure 9.5.6.1 Lemon sole in Subarea 4 and divisions 3.a and 7.d. Length-at-age data from Q1 (IBTS) and Q3 (IBTS & BTS) survey series, using data from all available years (2005–2024). To account for seasons, Q1 lengths are plotted at $a + 0.25$, Q3 lengths at $a + 0.75$. On the right: females only (the red line gives a fitted von Bertalanffy growth curve ($L_{\infty} = 299\text{ mm}$, $K = 0.356$, $t_0 = 0$)). On the left: males only (the red line gives a fitted von Bertalanffy growth curve ($L_{\infty} = 254\text{ mm}$, $K = 0.443$, $t_0 = 0$)).

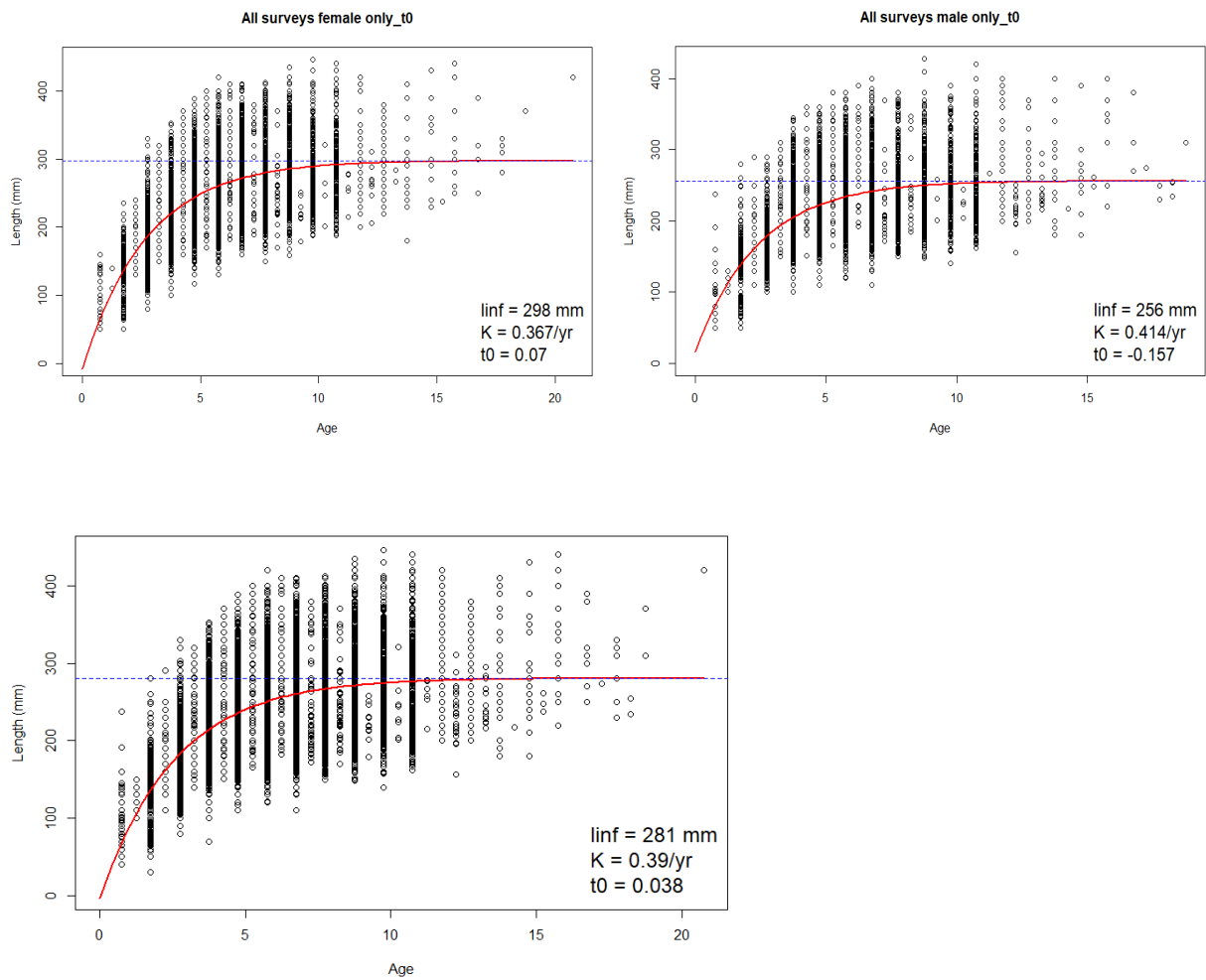


Figure 9.5.6.2 Lemon sole in Subarea 4 and divisions 3.a and 7.d. Length-at-age data from Q1 (IBTS) and Q3 (IBTS & BTS) survey series, using data from all available years (2005–2024). To account for seasons, Q1 lengths are plotted at $a + 0.25$, Q3 lengths at $a + 0.75$. On the top row: females (red line gives a fitted von Bertalanffy growth curve ($L_{\infty} = 298$ mm, $K = 0.367$, $t_0 = 0.07$). For males (red line gives a fitted von Bertalanffy growth curve ($L_{\infty} = 256$ mm, $K = 0.414$, $t_0 = -0.157$). Below: both females and males together (red line gives a fitted von Bertalanffy growth curve ($L_{\infty} = 281$ mm, $K = 0.39$, $t_0 = 0.038$).

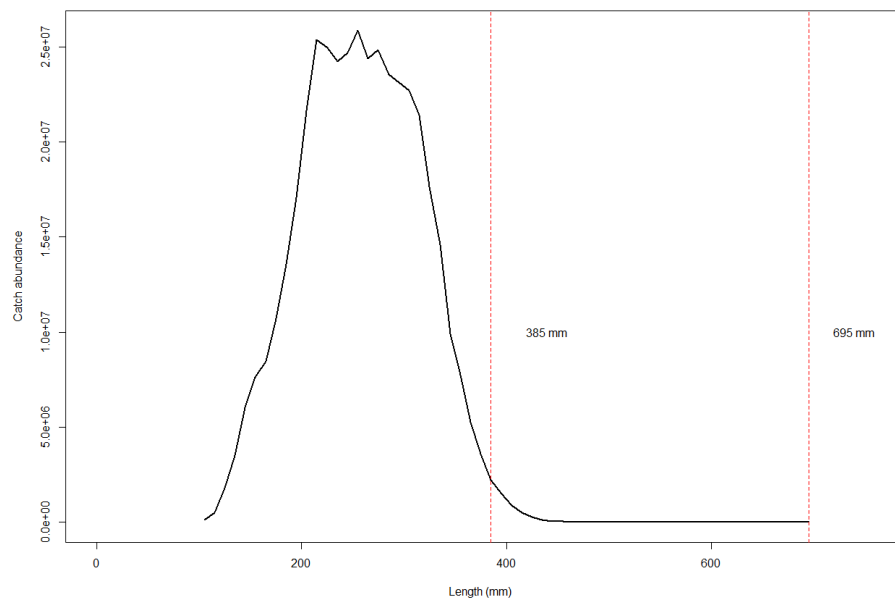


Figure 9.5.7. Lemon sole in Subarea 4 and divisions 3.a and 7.d. Length distribution of the commercial catch data submitted to InterCatch, collated over all available years (2002–2024). The red lines give (from left to right) the 99 percentile of the distribution (385 mm) and the longest observed fish (695 mm).

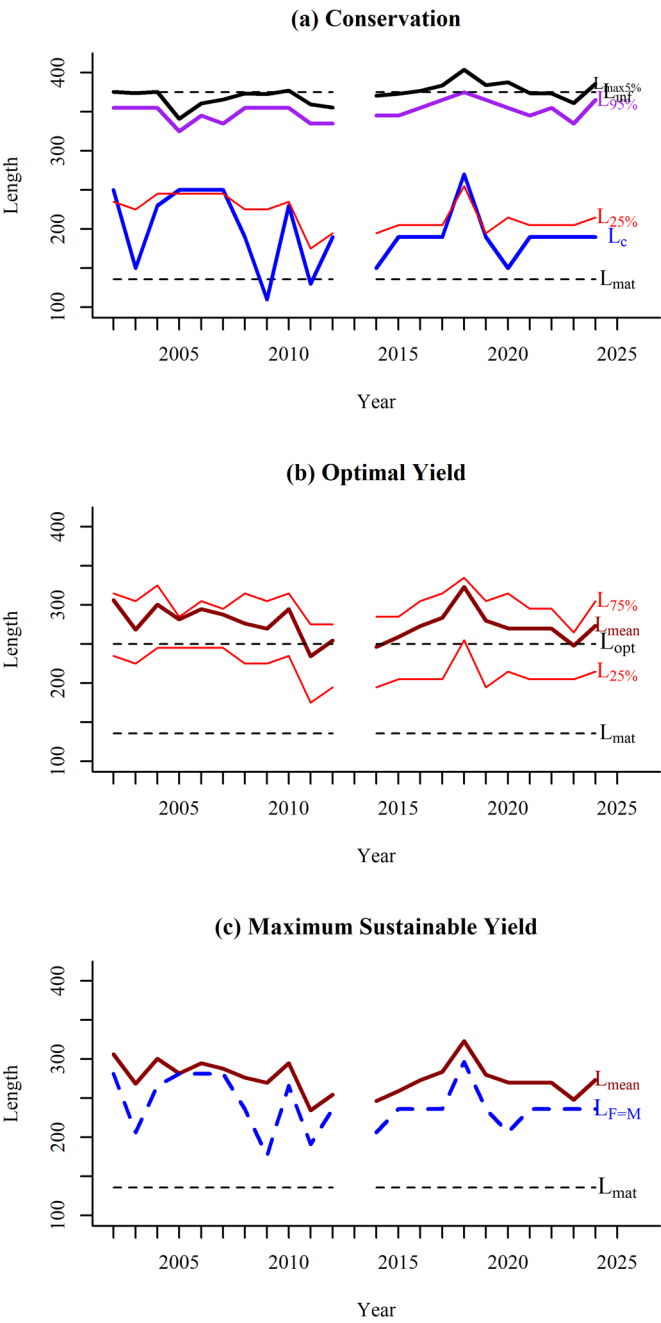


Figure 9.5.8. Lemon sole in Subarea 4 and divisions 3.a and 7.d. Results of LBI analysis (absolute estimates) using an L_{∞} =375mm.

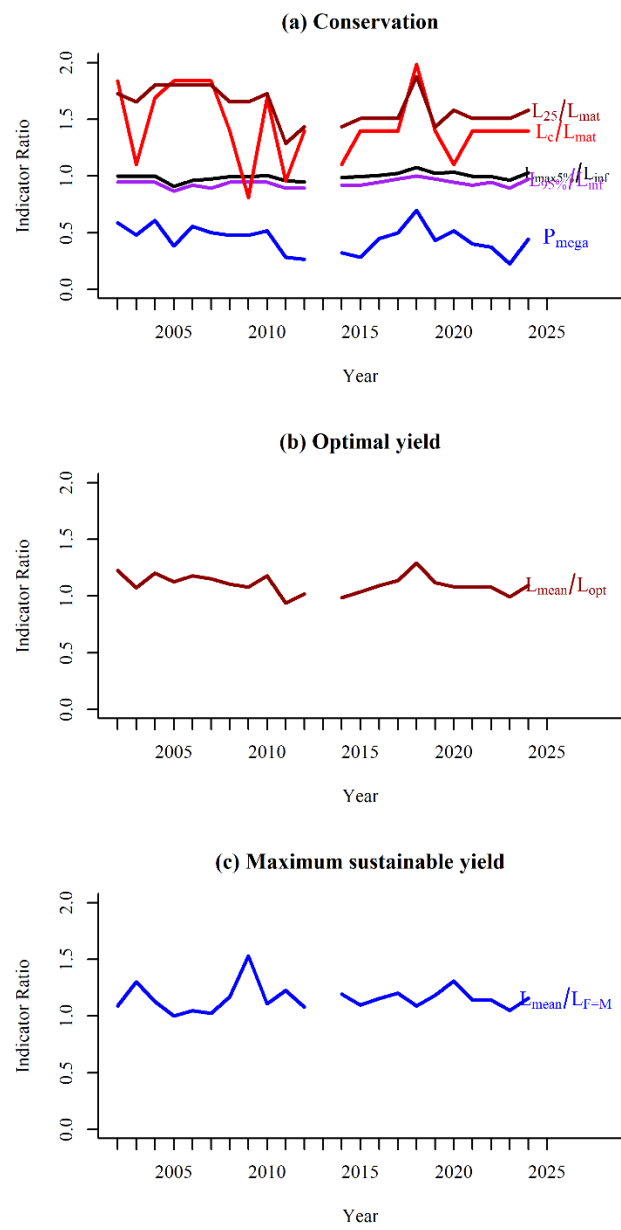


Figure 9.5.9. Lemon sole in Subarea 4 and divisions 3.a and 7.d. Results of LBI analysis (ratio estimates).

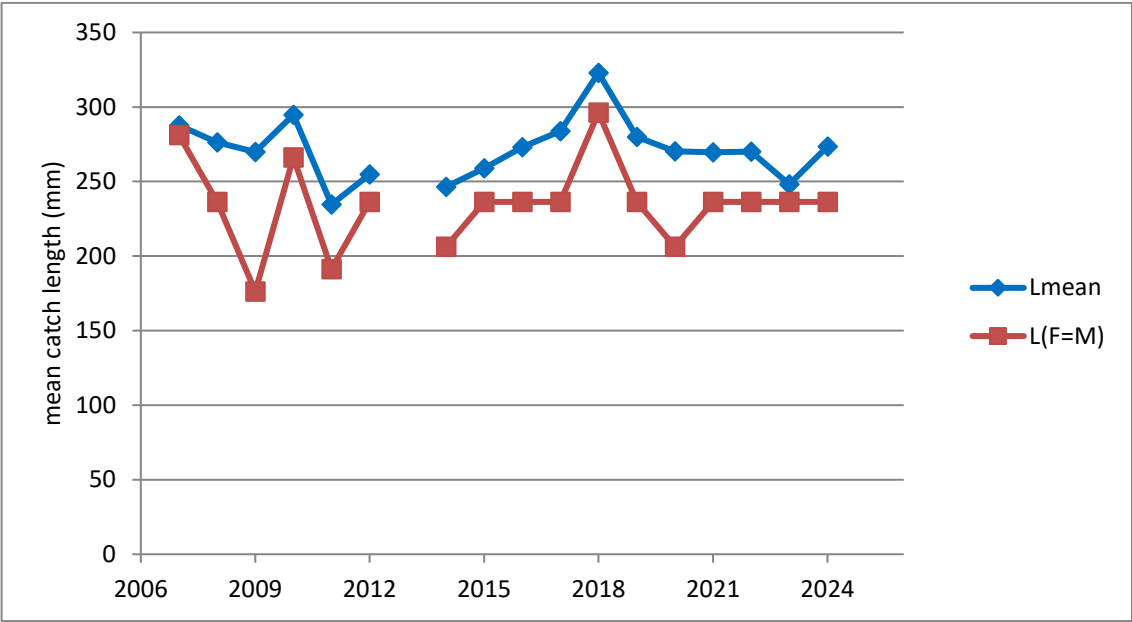


Figure 9.5.10. Lemon sole in Subarea 4 and divisions 3.a and 7.d. Comparison of L_{mean} (blue) and $L_{F=M}$ (red) for years 2007–2024 (excluding 2013 due to unlikely high Dutch discards), from LBI analysis.

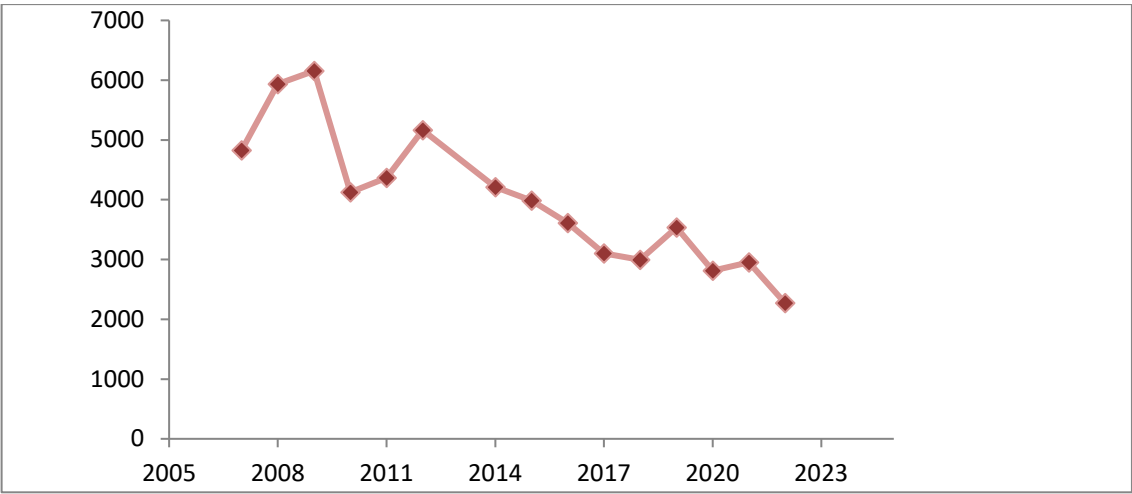


Figure 9.5.11. Lemon sole in Subarea 4 and divisions 3.a and 7.d. Harvest rate, ratio of catch C to stock size indicator I for years 2007–2022. The year 2013 was removed due to unlikely high Dutch discards. The 2023 and 2024 harvest rate were not used.

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